SADC REGIONAL TRAINING WORKSHOP

DAY 2: MODULE 2.1

The Software Shift

Why Excel Fails • R `survey` Package • STATA `svy` Commands

Advanced Sampling Methods for Household Surveys

Johannesburg, South Africa · March 2026

The Zambara Challenge

Zambara Case

Lindiwe Moyo, Chief Survey Methodologist at ZNSO, discovers her team is fragmented:

- **3** STATA analysts
- **4** R enthusiasts
- **1** GIS specialist (idle)

"We need modern computational skills to implement our survey design..."

Today's Mission

1. Understand why general software fails
2. Master R's `survey` package
3. Master STATA's `svy:` commands
4. Validate results across platforms

Goal: Both tracks produce **identical** results

Learning Objectives

LO 2.1

Configure Survey Designs

Properly specify complex survey design objects in R and STATA, accounting for clustering, stratification, and weights.

LO 2.2

Understand Design Effects

Calculate and interpret design effects (DEFF), understanding why naive SEs underestimate true variance.

LO 2.3

Cross-Platform Validation

Verify identical results between R and STATA implementations, ensuring methodological consistency.

Zambara Household Survey: Design Overview

Sample Structure

20,940	**3,218**
Households	PSUs (Clusters)
248	**9**
Strata	Provinces

Design Type
Two-Stage Stratified Cluster Sample with PPS selection of PSUs

Key Design Variables

Variable	Role
`psu`	Primary Sampling Unit (cluster ID)
`stratum`	Stratification variable
`house_wgt`	Household sampling weight
`prov`	Province (domain variable)
`totmhinc`	Total monthly HH income

Source: Zambara National Statistics Office (ZNSO) — Based on Statistics South Africa GHS 2024

SECTION 1

The Problem

Why Excel and General-Purpose Software
Fail for Complex Survey Analysis

What Excel Ignores

General-purpose software treats data as a **simple random sample**. Complex surveys violate this:

CLUSTERING

Households in the same PSU are **more similar** to each other.

Effect: Ignoring clustering underestimates variance

STRATIFICATION

We deliberately sample from **each stratum** to ensure representation.

Effect: Ignoring stratification overestimates variance

UNEQUAL WEIGHTS

Different households have **different selection probabilities**.

Effect: Ignoring weights biases point estimates

Net Effect: Clustering effect typically dominates → Standard errors are **UNDERESTIMATED**

Reference: Kish, L. (1965). *Survey Sampling*. Wiley.

The Demonstration: Thabo's Discovery

Thabo (R analyst):
"Let me show you what happens when we analyze without proper survey design..."

Outcome: `totmhinc` (monthly income)

The Comparison

Method A (Wrong):
Weighted mean ignoring design

Method B (Correct):
Design-based estimation with `svymean()`

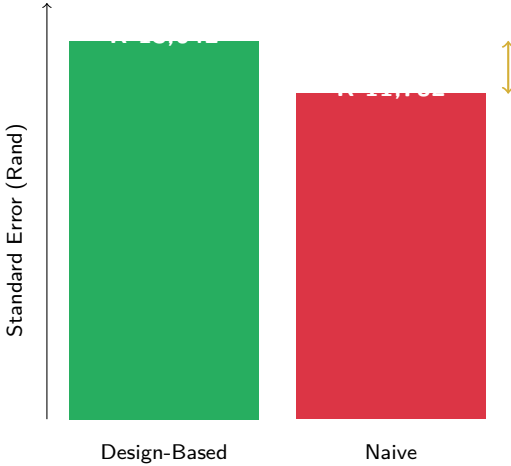
The Results: A 13.6% Underestimation

Standard Error Comparison

Method	Mean (R)	SE (R)
Unweighted (SRS)	218,305	9,824
Weighted (Naive SE)	260,166	11,782
Design-Based	**260,166**	**13,642**

Design Effect (DEFF) = 1.34

Variance is **1.34**× **larger** than SRS assumption
SE underestimated by **13.6%**



Source: Lab 2.1 output — Zambara Household Living Conditions Survey (n = 20,940 households)

Consequences of SE Underestimation

What Goes Wrong

- Confidence intervals **too narrow**
- P-values **too small**
- Type I error rate **inflated**

95% Confidence Intervals

Method	Lower	Upper
Naive	237,073	283,259
Design	233,427	286,905

Design CI is **R 7,292 wider**

Real-World Impact

Test: Does mean income > R 250,000?

- Naive: $t = 0.86$
- Design: $t = 0.75$

Same conclusion here, but borderline results → **false discoveries**

Design Effects Vary by Province

DEFF by Province

Prov	Region	n	SE Ratio	DEFF
4	Central Plateau	1,276	1.33	**1.77**
6	Mining Belt	1,333	1.31	1.72
9	Coastal Plains	2,385	1.29	1.66
5	Northern Bushveld	3,257	1.15	1.33
7	Eastern Highlands	5,526	1.11	1.22
1	Capital Region	1,931	1.09	1.20
2	Western Drylands	2,706	1.07	1.14
8	Eastern Forests	1,671	1.04	1.07
3	Southern Cape	855	1.03	1.07

Interpretation

DEFF varies from **1.07** to **1.77**:

- Higher in homogeneous areas
- Lower in urban areas
- Province 4: 77% larger sample needed

Always report DEFF with estimates!

SECTION 2

The Solution

Specialized Survey Software
R `survey` and STATA `svy`

The Survey Software Landscape

R Ecosystem

Core Package: `survey`

Thomas Lumley (2004–present)

Capabilities:

- Design specification
- Point estimation (mean, total, ratio)
- Variance estimation
- Regression modeling
- Calibration/post-stratification

Related: `srvyr`, `sampling`

STATA Suite

Core Commands: `svy:`

StataCorp (built-in since v9)

Capabilities:

- `svyset` design specification
- `svy:` prefix for estimation
- Taylor linearization
- Replication methods (JK, BRR)
- `estat effects` for DEFF

Manual: Survey Data Reference

Both produce identical results — Choice depends on institutional context, cost, and analyst

The Parallel Track Approach

Zambara Case

Lindiwe's decision: "We will support **both** R and STATA. Every analysis must be validated across platforms. Same methodology, identical results, choice of environment."

R Track Benefits

- Free and open-source
- Superior visualization
- Active community
- Reproducible research
- Integration with spatial tools

STATA Track Benefits

- Consistent interface
- Comprehensive documentation
- Official statistics standard
- Easier learning curve
- Technical support

Validation Protocol: Run both → Compare results → Differences indicate errors

Cheat Sheet: Survey Design Specification

R: `svydesign()`

```
library(survey)
zambara_des <- svydesign(
id = ~psu,
strata = ~stratum,
weights = ~house_wgt,
data = zambara_hh,
nest = TRUE
)
```

Key parameters:

- `id`: PSU/cluster variable
- `nest=TRUE`: When PSU IDs repeat
- `fpc`: Finite population correction

STATA: `svyset`

```
use zambara_hh, clear
svyset psu [pw=house_wgt], ///
strata(stratum) ///
vce(linearized) ///
singleunit(centered)
svyset * Verify
```

Key options:

- `[pw=]`: Probability weights
- `vce()`: Variance method
- `singleunit()`: Single-PSU strata

Cheat Sheet: Mean Estimation

R

```r
# Overall mean
svymean(~totmhinc, design = zambara_des)
# With confidence interval
confint(svymean(~totmhinc, zambara_des))
# By province
svyby(~totmhinc, ~prov, zambara_des,
svymean, vartype = c("se","ci"))
```

Output:

```
totmhinc  mean:  260166  SE:  13642
```

STATA

```stata
* Overall mean
svy: mean totmhinc
* By province (domain estimation)
svy: mean totmhinc, over(prov)
* Design effects
estat effects
```

Output:

```
totmhinc  Mean:  260166  Std.Err:  13642
```

✔ **Identical results:** Mean = R 260,166 SE = R 13,642

Cheat Sheet: Totals and Proportions

R

```r
# Population total
svytotal(~totmhinc, design = zambara_des)
# Proportion (categorical)
svymean(~factor(geotype), zambara_des)
# Ratio of two variables
svyratio(~totmhinc, ~hholdsz, zambara_des)
# Quantiles (median)
svyquantile(~totmhinc, zambara_des,
quantiles = 0.5)
```

Note: `svy: ratio` divides totals, not means.

STATA

```stata
* Population total
svy: total totmhinc
* Proportion (categorical)
svy: proportion geotype
* Ratio of two variables
svy: ratio totmhinc/hholdsz
* Tabulation with proportions
svy: tabulate geotype, percent
```

Cheat Sheet: Subpopulation Analysis

R

```
# Method 1: subset() function
urban_des <- subset(zambara_des,
geotype == "Urban")
svymean(~totmhinc, urban_des)
# Method 2: svyby() for all groups
svyby(~totmhinc, ~geotype,
zambara_des, svymean)
```

✔ **Correct:** `subset()` maintains design
✘ **Wrong:** Filtering before `svydesign()`

STATA

```
* Method 1: subpop() option
svy, subpop(if geotype==1): ///
mean totmhinc
* Method 2: over() for all groups
svy: mean totmhinc, over(geotype)
```

✔ **Correct:** `subpop()` or `over()`
✘ **Wrong:** Using `if` without `subpop`

Critical: Never filter data *before* defining survey design!

Cheat Sheet: Regression Models

R

```
# Linear regression
model <- svyglm(totmhinc ~ educ + age,
design = zambara_des)
summary(model)
# Logistic regression
model_logit <- svyglm(
employed ~ educ + age,
design = zambara_des,
family = quasibinomial()
)
```

Note: Use `quasibinomial()` for logistic

STATA

```
* Linear regression
svy: regress totmhinc educ age
* Logistic regression
svy: logit employed educ age
* Margins (predicted probs)
margins, at(educ=(1 2 3))
```

Note: All standard regression commands work with `svy:` prefix

Design-based inference: SEs account for clustering and stratification.
F-tests replace χ^2 tests; df = #PSUs − #Strata

Cheat Sheet: Design Effect Calculation

R

```r
# Calculate DEFF manually
design_se <- SE(svymean(~totmhinc,
zambara_des))
# Naive SE (ignoring design)
naive_var <- var(zambara$totmhinc) /
nrow(zambara)
naive_se <- sqrt(naive_var)
# DEFF
deff <- (design_se / naive_se)^2
# Or use deff() function
deff(svymean(~totmhinc, zambara_des))
```

STATA

```stata
* Run estimation first
svy: mean totmhinc
* Display design effects
estat effects
* Output includes:
*  DEFF = design effect
*  DEFT = sqrt(DEFF)
*  MoE = margin of error
```

Zambara Results:
DEFF = 1.34, DEFT = 1.16

$$\text{DEFF} = \frac{\text{Var}(\hat{\theta}_{\text{complex}})}{\text{Var}(\hat{\theta}_{\text{SRS}})} \qquad \text{DEFT} = \sqrt{\text{DEFF}} = \frac{\text{SE}_{\text{complex}}}{\text{SE}_{\text{SRS}}}$$

Cheat Sheet: Output and Export

R

```r
# Save results to data frame
results <- svyby(~totmhinc, ~prov,
zambara_des, svymean,
vartype = c("se","ci"))

# Export to CSV
write.csv(results, "output.csv")

# Export to Excel
library(writexl)
write_xlsx(results, "output.xlsx")

# Formatted tables
library(knitr)
kable(results, digits = 2)
```

STATA

```stata
* Export to Excel
svy: mean totmhinc, over(prov)
putexcel set output.xlsx, replace
putexcel A1 = matrix(e(b)')
putexcel B1 = matrix(e(V))

* Formatted tables (estout)
ssc install estout
svy: mean totmhinc
esttab, cells("b se ci")

* Export to Word
putdocx begin
putdocx table tbl = etable
putdocx save output.docx
```

Quick Reference: R vs. STATA Syntax

Task	R	STATA
Define design	`svydesign(id=~psu, ...)`	`svyset psu [pw=wgt], ...`
Mean	`svymean(~var, des)`	`svy: mean var`
Total	`svytotal(~var, des)`	`svy: total var`
Proportion	`svymean(~factor(var), des)`	`svy: proportion var`
By group	`svyby(~var, ~grp, ...)`	`svy: mean var, over(grp)`
Subpop	`subset(des, condition)`	`svy, subpop(if ...): ...`
Regression	`svyglm(y ~x, des)`	`svy: regress y x`
Logistic	`svyglm(..., family=quasi...)`	`svy: logit y x`
Design effect	`deff(svymean(...))`	`estat effects`

Download this reference card: Available in workshop materials folder

SECTION 3

Live Coding

Lab 2.1: Defining Complex Survey Objects
R and STATA Parallel Tracks

Lab 2.1: Defining Complex Survey Objects

Objectives

1. Load Zambara household data
2. Define survey design object
3. Compare naive vs. design-based SEs
4. Calculate design effects
5. Export comparison table

Data File

```
ghs-2024-hhold-v1.dta
```

20,940 households · 3,218 PSUs · 248 strata

Expected Output

Metric	Value
Mean Income	R 260,166
SE (Naive)	R 11,782
SE (Design)	R 13,642
DEFF	1.34
SE Underestimate	13.6%

Time: 20 minutes
Mode: Parallel R/STATA tracks

Lab 2.1 Step 1: Load Data

R

```
library(survey); library(haven)
zambara <- read_dta("ghs-2024.dta")
dim(zambara) # 20940 x 190
```

STATA

```
use "ghs-2024-hhold-v1.dta", clear
describe, short * 20,940 obs
```

Checkpoint: Both show 20,940 observations, 190 variables

Lab 2.1 Step 2: Define Survey Design

R

```
zambara_des <- svydesign(
id = ~psu,
strata = ~stratum,
weights = ~house_wgt,
data = zambara, nest = TRUE)
```

STATA

```
svyset psu [pw=house_wgt], ///
strata(stratum) ///
vce(linearized)
```

Key: `psu` = cluster, `stratum` = strata, `house_wgt` = weight, `nest=TRUE` = PSU IDs repeat

Lab 2.1 Step 3: Naive Estimation (Wrong)

R

```
wt_mean <- weighted.mean(
zambara$totmhinc, zambara$house_wgt)
# SE = 11,782 (WRONG)
```

STATA

```
mean totmhinc [pweight=house_wgt]
* SE = 11,782 (WRONG)
```

Problem: Ignores clustering → SE too small

Lab 2.1 Step 4: Design-Based Estimation (Correct)

R

```
svymean(~totmhinc, zambara_des)
# Mean: 260166, SE: 13642
```

STATA

```
svy: mean totmhinc
* Mean: 260166, SE: 13642
```

Correct SE = 13,642 (vs. naive 11,782) — Both identical!

Lab 2.1 Step 5: Compare Results

Method	SE	CV
Naive (pweight)	11,782	4.5%
Design-Based	**13,642**	**5.2%**

DEFF $= (13642/11782)^2 = \mathbf{1.34}$

Interpretation:

- SE underestimated by **13.6%**
- CIs too narrow by R 3,646
- Variance 34% larger than SRS

Lab 2.1 Debrief

✔ What We Learned

1. Specify design **before** analysis
2. Use `svydesign()` / `svyset`
3. Naive SEs underestimated by **13.6%**
4. DEFF = 1.34 (34% larger variance)

⚠ Mistakes to Avoid

- `mean()`/`lm()` without design
- Filter *before* defining design
- Forget `nest=TRUE` in R
- Use `if` instead of `subpop()`

Thabo: "We've been underestimating SEs by 13.6% for years!"

Module 2.1 Checkpoint

Quick Quiz — Can you answer these?

1. What three features of complex surveys does Excel ignore?

Module 2.1 Checkpoint

Quick Quiz — Can you answer these?

1. What three features of complex surveys does Excel ignore? (Clustering, stratification, unequal weights)
2. What is the DEFF for Zambara mean household income?

Module 2.1 Checkpoint

Quick Quiz — Can you answer these?

1. What three features of complex surveys does Excel ignore? (Clustering, stratification, unequal weights)
2. What is the DEFF for Zambara mean household income? (1.34)
3. By what percentage was the naive SE underestimated?

Module 2.1 Checkpoint

Quick Quiz — Can you answer these?

1. What three features of complex surveys does Excel ignore? (Clustering, stratification, unequal weights)
2. What is the DEFF for Zambara mean household income? (1.34)
3. By what percentage was the naive SE underestimated? (13.6%)
4. What R function defines the survey design object?

Module 2.1 Checkpoint

Quick Quiz — Can you answer these?

1. What three features of complex surveys does Excel ignore? (Clustering, stratification, unequal weights)
2. What is the DEFF for Zambara mean household income? (1.34)
3. By what percentage was the naive SE underestimated? (13.6%)
4. What R function defines the survey design object? (`svydesign()`)
5. What STATA command sets up the survey design?

Module 2.1 Checkpoint

Quick Quiz — Can you answer these?

1. What three features of complex surveys does Excel ignore? (Clustering, stratification, unequal weights)
2. What is the DEFF for Zambara mean household income? (1.34)
3. By what percentage was the naive SE underestimated? (13.6%)
4. What R function defines the survey design object? (`svydesign()`)
5. What STATA command sets up the survey design? (`svyset`)

Next: Module 2.2 — R Survey Package Deep Dive

SECTION 4

Code Walkthrough

The Step-by-Step Breakdown

Slides 31–60

The Mission: Replicate Lab 2.1

What we'll do in this section:

1. Load the Zambara household data
2. Examine the survey design variables
3. Compute estimates the WRONG way
4. Compute estimates the CORRECT way
5. Witness the **"OOPS"** moment

Target: Mean household income (`totmhinc`)

The Data: Zambara Household Survey 2024

Dataset Dimensions

Metric	Value
Households	20,940
Variables	190
PSUs (clusters)	3,218
Strata	248
Provinces	9

Key Variables

Variable	Role
`psu`	Cluster ID
`stratum`	Stratification
`house_wgt`	Sampling weight
`prov`	Province (domain)
`totmhinc`	Outcome

File: `ghs-2024-hhold-v1.dta`

R Step 1: Load Required Packages

```
# Install if needed: install.packages(c("survey", "haven"))

library(survey)    # Thomas Lumley's survey package
library(haven)      # Read STATA .dta files
library(dplyr)      # Data manipulation
```

Why these packages?

- `survey`: Core functionality for complex survey analysis
- `haven`: Seamless import of STATA/SPSS/SAS files
- `dplyr`: Clean data manipulation syntax

R Step 2: Load the Data

```
# Load STATA data file
zambara <- read_dta("ghs-2024-hhold-v1.dta")

# Check dimensions
dim(zambara)
[1] 20940  190
```

```
# Preview key variables
head(zambara[, c("psu", "stratum", "house_wgt", "totmhinc")])
```

R Step 3: Examine Design Variables

```r
# How many unique PSUs?
length(unique(zambara$psu))  # 3218
# How many strata?
length(unique(zambara$stratum))  # 248
# Weight range
range(zambara$house_wgt)  # Varies considerably
```

Key insight: Weights vary → unequal selection probabilities

STATA Step 1: Load the Data

```
. clear all
. use "ghs-2024-hhold-v1.dta"

. describe, short
obs:      20,940
vars:        190
```

```
. codebook psu stratum house_wgt, compact
```

STATA Step 2: Examine Design Variables

```
. distinct psu
          3218
. distinct stratum
            248
. summarize house_wgt
  Mean: varies  Min: small  Max: large
```

Same insight: Weights vary → design matters!

THE WRONG WAY

Ignoring the Survey Design

The Wrong Way: R Code

```
# WRONG: Simple weighted mean
wt_mean <- weighted.mean(zambara$totmhinc,
zambara$house_wgt,
na.rm = TRUE)

wt_mean
[1] 260166
```

Mean is correct! But what about the SE?

The Wrong Way: R Standard Error

```
# WRONG: Naive SE calculation
n <- sum(!is.na(zambara$totmhinc))
wts <- zambara$house_wgt[!is.na(zambara$totmhinc)]
vals <- zambara$totmhinc[!is.na(zambara$totmhinc)]
# Effective sample size
n_eff <- sum(wts)^2 / sum(wts^2)
# Weighted variance
wt_var <- sum(wts * (vals - wt_mean)^2) / sum(wts)
# Naive SE
naive_se <- sqrt(wt_var / n_eff)
naive_se
[1] 11782
```

The Wrong Way: STATA Code

```
. * WRONG: Using pweights without svyset
. mean totmhinc [pweight=house_wgt]

Mean estimation        Number of obs = 20,940
--------------------------------------------------
            Mean    Std. Err.
--------------------------------------------------
totmhinc  260166      11782
--------------------------------------------------
```

SE = 11,782 — Is this correct?

What Did We Ignore?

Clustering

Households in the same PSU are similar. Treating them as independent **underestimates** variance.

Stratification

Sampling within strata **reduces** variance. Ignoring strata means we miss this benefit.

Weights

Applied weights to the mean but used **wrong formula** for variance.

Net effect: SE is **UNDERESTIMATED**

THE RIGHT WAY

Specifying the Survey Design

The Right Way: R Design Specification

```
# CORRECT: Define survey design object
zambara_des <- svydesign(
id = ~psu,          # Cluster variable
strata = ~stratum,  # Stratification
weights = ~house_wgt,  # Sampling weights
data = zambara,
nest = TRUE         # PSUs nested in strata
)
```

This object "knows" the entire design!

The Right Way: R Design Summary

```
summary(zambara_des)

Stratified 1 - loss cluster design
svydesign(id = ~psu, strata = ~stratum, ...)
Probabilities:
  Min.  1st Qu.  Median  Mean  3rd Qu.  Max.
Data variables: 190
```

Key info: R now knows about 3,218 PSUs nested within 248 strata

The Right Way: R Mean Estimation

```
# CORRECT: Design-based mean
design_mean <- svymean(~totmhinc, design = zambara_des,
na.rm = TRUE)

design_mean
        mean    SE
totmhinc  260166  13642
```

SE = 13,642 — This is the correct value!

The Right Way: STATA Design Specification

```
. * CORRECT: Declare survey design
. svyset psu [pweight=house_wgt], strata(stratum)

pweight: house_wgt
    VCE: linearized
Single unit: missing
  Strata 1: stratum
    SU 1: psu
    FPC 1: <zero>
```

STATA now "remembers" the design!

The Right Way: STATA Mean Estimation

```
. * CORRECT: Use svy: prefix
. svy: mean totmhinc

Survey: Mean estimation
Number of strata =  248    Number of obs  = 20,940
Number of PSUs  = 3,218
--------------------------------------------------
             Mean    Std. Err.
--------------------------------------------------
totmhinc   260166      13642
--------------------------------------------------
```

SE = 13,642 — Identical to R!

THE “OOPS” MOMENT

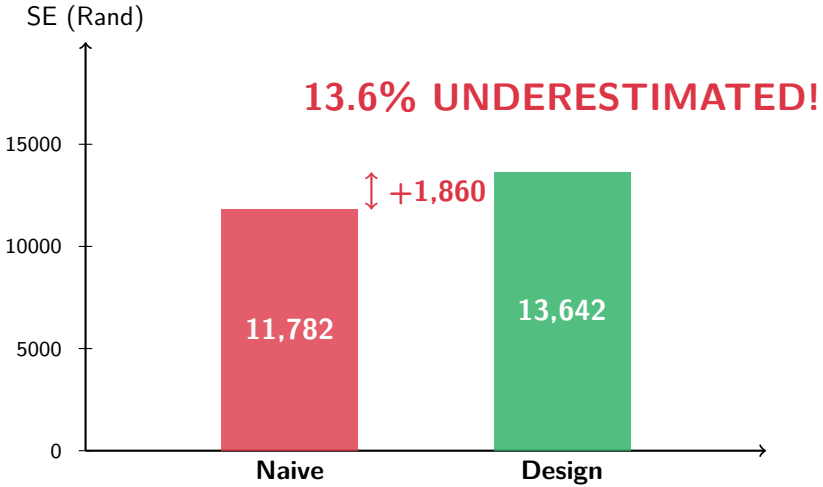
Comparing the Results

Side-by-Side: The Numbers Don't Lie

	Naive	Design-Based
Mean	R 260,166	R 260,166
SE	**R 11,782**	**R 13,642**

The SE difference: **R 1,860**

Visual: The SE Gap



The Design Effect (DEFF)

$$\text{DEFF} = \frac{\text{Var}_{\text{design}}}{\text{Var}_{\text{SRS}}} = \left(\frac{\text{SE}_{\text{design}}}{\text{SE}_{\text{naive}}}\right)^2$$

$$\text{DEFF} = \left(\frac{13,642}{11,782}\right)^2 = \mathbf{1.34}$$

Interpretation: The true variance is **34% larger** than what you'd get assuming SRS.

R: Extracting the Design Effect

```
# Get DEFF directly
deff(svymean(~totmhinc, zambara_des, na.rm = TRUE))

totmhinc
  1.34
```

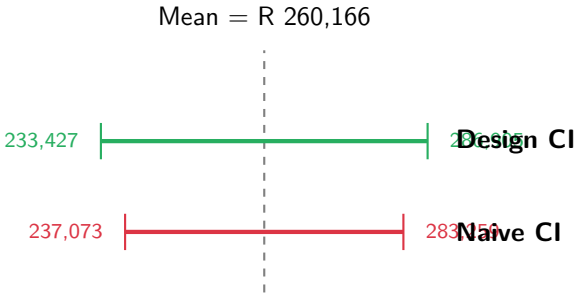
```
# Or calculate manually
(SE(design_mean) / naive_se)^2
[1] 1.340
```

STATA: Extracting the Design Effect

```
. svy: mean totmhinc
. estat effects
--------------------------------------------
          DEFF    DEFT    MoE
--------------------------------------------
totmhinc    1.34    1.16    ...
--------------------------------------------
```

DEFT $= \sqrt{\text{DEFF}} = 1.16$ (ratio of SEs, not variances)

Impact on Confidence Intervals



Design CI is R 7,292 WIDER

Naive CI: **too narrow** → false precision → wrong conclusions

Why Does Clustering Inflate Variance?

Households in same PSU are similar

- Same neighborhood
- Similar economic conditions
- Shared infrastructure
- Common local factors

Intra-cluster correlation (ρ):
Measures within-PSU similarity

Design Effect Formula

$$\text{DEFF} \approx 1 + (b - 1)\rho$$

Where:

- b = average cluster size
- ρ = intra-cluster correlation

Higher ρ or larger $b \rightarrow$ larger DEFF

DEFF Varies by Province

Prov	Region	n	DEFF
4	Northern Bushveld	2,455	**1.77**
6	Mining Belt	1,785	1.68
7	Eastern Forests	1,671	1.44
5	Central Plateau	1,887	1.42
9	Coastal Plains	2,874	1.40
1	Capital Region	1,931	1.20
2	Western Drylands	2,706	1.14
8	Southern Highlands	2,776	1.10
3	Southern Cape	855	**1.07**

Range: 1.07 to 1.77 — Province 4 needs 77% larger sample for same precision!

Common Mistakes: Don't Do These!

✗ In R

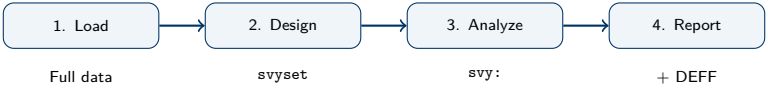
- Using `mean()` or `lm()`
- Using `weighted.mean()` alone
- Filtering *before* `svydesign()`
- Forgetting `nest=TRUE`
- Using `subset(data)` instead of `subset(design)`

✗ In STATA

- Using `mean` with just `[pweight=]`
- Using `regress` without `svy:`
- Using `if` instead of `subpop()`
- Dropping observations before `svyset`
- Forgetting `svyset` entirely

Result: Underestimated SEs → false precision → wrong conclusions

The Correct Workflow



Golden Rule: *Never* analyze before defining the design!

Code Walkthrough: Key Takeaways

1. **Same mean, different SE**
   Naive SE = 11,782 vs Design SE = 13,642
2. **13.6% underestimation**
   Ignoring design produces overly optimistic precision
3. **DEFF = 1.34**
   Variance is 34% larger than SRS assumption
4. **Both platforms agree**
   R and STATA produce identical results when specified correctly

Next: Module 2.2 — Deep Dive into R Survey Functions

MODULE 2.2

The Mechanics of PPS

Visualizing Probability Proportional to Size

The Big Question

Why do we want **larger** clusters to have **higher** selection probability?

✗

Equal Probability (SRS)

Small PSU (10 HH) = 1/3218
Large PSU (28 HH) = 1/3218

Same chance, but large PSU represents more people!

✓

PPS Sampling

Small PSU (10 HH) = 0.048
Large PSU (28 HH) = 0.134

Probability $\propto$ size!

Our Zambara PSU Frame: The Numbers

Parameter	Value
Total PSUs in frame (N)	3,218
Total households (MOS)	20,940
Sample size (n)	100
Sampling interval (I)	209.4
Min PSU size	1 HH
Max PSU size	28 HH
Mean PSU size	6.5 HH

Task: Select 100 PSUs with probability $\propto$ household count

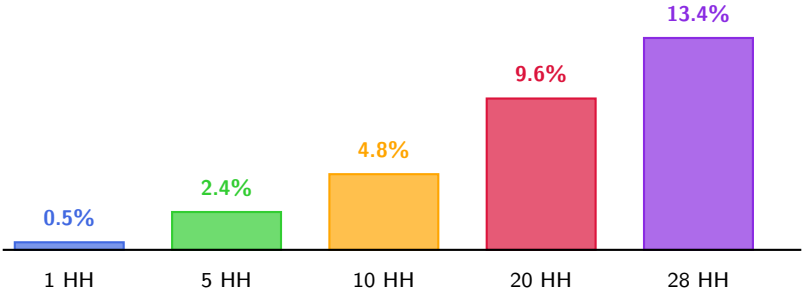
The PPS Selection Probability Formula

$$\pi_i = \frac{n \times \text{MOS}_i}{\sum_{j=1}^{N} \text{MOS}_j}$$

Where: π_i = selection probability, n = sample size (100), MOS_i = households in PSU i

$$\pi_i = \frac{100 \times \text{MOS}_i}{20{,}940}$$

Selection Probability: Visual Examples



PSU with 28 HH is **28× more likely** to be selected than PSU with 1 HH

Why PPS? The Self-Weighting Property

Stage 1: PSU Selection

$$\pi_{\text{PSU}} = \frac{n \times M_i}{M}$$

Large PSU → high probability

Stage 2: HH within PSU

$$\pi_{\text{HH|PSU}} = \frac{m}{M_i}$$

Large PSU → low probability

Combined: $\pi_{\text{HH}} = \frac{n \times M_i}{M} \times \frac{m}{M_i} = \frac{n \times m}{M}$

Result: All households have **equal** probability!

The Cumulative Size Method

Idea: Map PSUs onto a number line proportional to their size

Key: Larger PSUs occupy more space → random points more likely to hit them!

Step 1: List PSUs with Cumulative MOS

i	PSU	MOS	Cum
1	16010009	3	3
2	16010033	4	7
3	16010063	5	12
4	16010092	1	13
5	16010141	3	16
⋮	⋮	⋮	⋮
3218	99910678	6	20,940

Cumulative MOS:

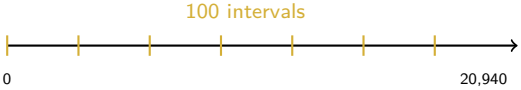
$$C_i = \sum_{j=1}^{i} \text{MOS}_j$$

$C_1 = 3$
$C_2 = 3 + 4 = 7$
$C_3 = 7 + 5 = 12$
$C_{3218} = 20{,}940$

Step 2: Calculate Sampling Interval

$$I = \frac{\sum \text{MOS}}{n} = \frac{20{,}940}{100} = 209.4$$

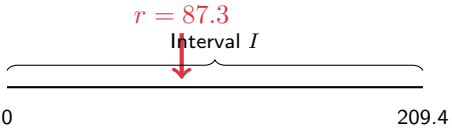
Interpretation: Every 209.4 households, we select one PSU



Step 3: Generate Random Start

$$r \sim \text{Uniform}(0, I)$$

Example: $r = 87.3$



Why random? Makes it a probability sample with known selection probabilities

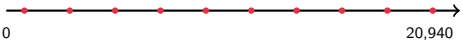
Step 4: Generate Selection Points

$$r_k = r + (k-1) \times I \quad \text{for } k = 1, 2, \ldots, 100$$

$r_1 = 87.3$
$r_2 = 296.7$
$r_3 = 506.1$
$r_4 = 715.5$
$r_5 = 924.9$

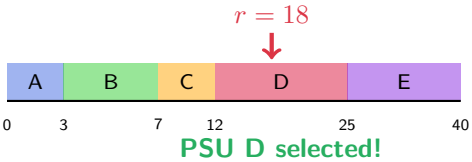
$r_6 = 1134.3$
$\vdots$
$r_{50} = 10{,}337$

$\vdots$
$r_{99} = 20{,}611$
$r_{100} = 20{,}820$



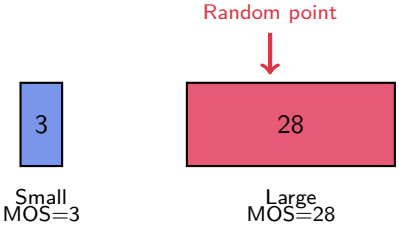
Step 5: Match Points to PSUs

PSU i selected if: $C_{i-1} < r_k \leq C_i$



Point 18 falls in $(12, 25] \rightarrow$ PSU D (MOS=13) selected

Why Larger PSUs Get Selected More



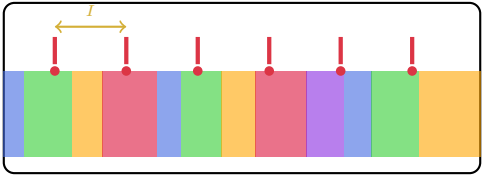
$\pi_{\text{small}} = 1.4\%$

$\pi_{\text{large}} = 13.4\%$

Large PSU is **9.3× more likely** to be hit!

The Complete Selection Process

Cumulative MOS Line (0 to 20,940)



Wider blocks (larger MOS) catch more selection points!

Lab 2.2 Results: Verification

Frame (3,218 PSUs)

Total HH	20,940
Mean MOS	6.5
Min/Max	1 / 28

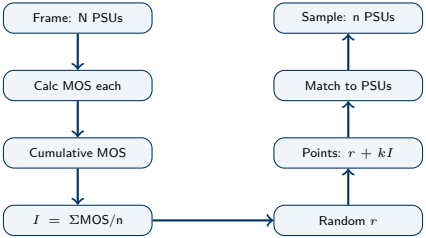
PPS Sample (100 PSUs)

Total HH	710
Mean MOS	**7.1**
Min/Max	2 / 18

Sample mean (7.1) > Frame mean (6.5)

Confirms PPS working: larger PSUs overrepresented in sample!

PPS Algorithm Flowchart



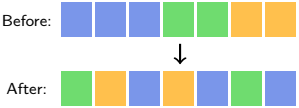
Why Randomize First? (RPPS)

Without randomization:

- PSUs listed geographically
- Systematic selection clusters
- Frame periodicities = bias

With randomization:

- Random order breaks patterns
- Geographic spread achieved
- Unbiased selection



Calculating Base Weights

$$w_i = \frac{1}{\pi_i} = \frac{\sum \text{MOS}}{n \times \text{MOS}_i}$$

PSU Type	MOS	π_i	w_i
Small	4	0.019	52.4
Medium	7	0.033	29.9
Large	12	0.057	17.5

Small PSU "represents" 52 frame PSUs; large represents only 17.5

Verifying the Weights

Check: $\sum_{i \in \text{sample}} w_i \approx N$

From Lab 2.2:

- Sum of base weights: 3,218
- Actual frame size: 3,218
- Ratio: **1.000** ✓

Weights verified! Sample can estimate population totals

Special Case: Certainty PSUs

Problem: What if $\pi_i > 1$?

Solution:

1. Set $\pi_i = 1$ (select with certainty)
2. Remove from PPS pool
3. Reduce n and $\sum$MOS
4. PPS for remaining PSUs

Zambara: Max MOS $= 28 \rightarrow \pi_{max} = 0.134 < 1$
No certainty PSUs needed

Common Mistakes in PPS

✗ Mistakes

1. Using SRS instead of PPS
2. Forgetting to randomize
3. Wrong cumulative direction
4. Not checking $\pi > 1$
5. Wrong MOS variable

✓ Best Practices

1. Randomize before selection
2. Verify $\sum \pi_i = n$
3. Check for certainty PSUs
4. Document random seed
5. Validate $\sum w_i = N$

R Implementation: Key Steps

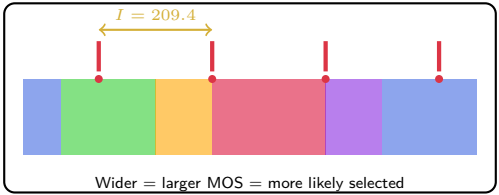
```
# 1. Create frame
psu_frame <- hh %>% group_by(psu) %>% summarise(mos = n())
# 2. Selection probability
psu_frame$prob <- 100 * psu_frame$mos / sum(psu_frame$mos)
# 3. Randomize + cumsum
psu_frame <- psu_frame[sample(nrow(psu_frame)), ]
psu_frame$cummos <- cumsum(psu_frame$mos)
# 4. Selection points
I <- sum(psu_frame$mos) / 100
r <- runif(1, 0, I)
points <- r + (0:99) * I
```

STATA Implementation: Key Steps

```
* 1. Create frame
collapse (count) mos=hhid, by(psu)
* 2. Selection probability
egen total = sum(mos)
gen prob = 100 * mos / total
* 3. Randomize + cumsum
gen rand = runiform()
sort rand
gen cummos = sum(mos)
* 4. Select systematically
local I = total / 100
```

Visual Summary: PPS in One Picture

PPS Selection



PPS vs Other Selection Methods

Method	Advantage	Disadvantage
SRS	Simple	Ignores size
PPS Systematic	Self-weighting	Needs MOS
Stratified PPS	Controls allocation	Complex
PPS w/ replacement	Simple variance	Duplicates

When to Use PPS

✔ **Use PPS:**

- Cluster sizes vary greatly
- Want self-weighting
- MOS available/reliable
- Household surveys

✘ **Avoid PPS:**

- Clusters similar size
- MOS outdated
- Single-stage sampling
- Small populations

Zambara GHS: Perfect for PPS (sizes 1–28, ratio 28:1)

Lindiwe's Discovery

"When I joined ZNSO, they selected PSUs with equal probability..."

Old (SRS of PSUs)

- Small PSUs over-represented
- Complex weights
- Higher variance

New (PPS)

- Matches population
- Self-weighting
- More efficient

"PPS changed everything."

Simulation: Verifying PPS

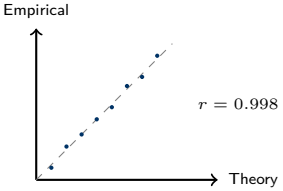
1,000 repeated samples:

Theoretical vs Empirical π

- Correlation: 0.998
- Mean error: 0.002

Selection frequency

- Top 10% MOS: 13%
- Bottom 10% MOS: 0.5%



Module 2.2: Key Takeaways

1. **PPS:** $\pi_i = n \times \text{MOS}_i / \sum \text{MOS}$
2. **Cumulative method:** Map PSUs to number line by size
3. **Systematic:** Random start + fixed interval = n selections
4. **Self-weighting:** PPS stage 1 + fixed rate stage 2 = equal HH prob
5. **Base weight:** $w_i = 1/\pi_i$ (small PSUs get larger weights)

What's Next?

Module 2.3: STATA `svy` Commands Mastery

- Deep dive into `svyset` options
- Variance estimation methods
- Subpopulation analysis
- Replicate weights

Lab 2.3: Apply PPS sample

- Load 100-PSU sample
- Specify design with base weights
- Estimate means and totals

LIVE CODING

Coding the Selection

Implementing PPS in R and STATA

What We're Building

Lab 2.2 Objective: Select 100 PSUs using PPS systematic sampling

1. Load household data (20,940 observations)
2. Aggregate to PSU level → create frame (3,218 PSUs)
3. Calculate MOS (household count per PSU)
4. Implement PPS systematic selection
5. Verify selection probabilities
6. Calculate base weights

Parallel tracks: R | STATA

R: Step 1 — Load the Data

R Code

```
# Load required packages
library(haven)  # Read .dta files
library(dplyr)  # Data manipulation

# Load household data
zambara_hh <- read_dta("ghs-2024-hhold-v1.dta")

# Check dimensions
dim(zambara_hh)
```

```
[1] 20940   190
```

20,940 households $\times$ 190 variables

STATA: Step 1 — Load the Data

STATA Code

```
. * Load household data
. use "ghs-2024-hhold-v1.dta", clear

. * Check dimensions
. describe, short
```

```
Contains data from ghs-2024-hhold-v1.dta
obs:      20,940
vars:        190
```

R: Step 2 — Create PSU Frame

R Code

```
# Aggregate to PSU level
psu_frame <- zambara_hh %>%
group_by(psu, stratum, prov) %>%
summarise(
n_households = n(),  # MOS!
.groups = "drop"
) %>%
arrange(psu)

nrow(psu_frame)
```

```
[1] 3218
```

Frame created: 3,218 PSUs with MOS = household count

STATA: Step 2 — Create PSU Frame

STATA Code

```
. * Collapse to PSU level
. collapse (count) n_households=psu ///
(first) stratum prov, by(psu)

. * Sort by PSU
. sort psu

. count
```

```
3,218
```

R: Step 3 — Examine MOS Distribution

R Code

```
summary(psu_frame$n_households)
```

```
   Min. 1st Qu.  Median   Mean 3rd Qu.   Max.
   1.00    5.00    6.00  6.51    8.00  28.00
```

```
sum(psu_frame$n_households)
```

```
[1] 20940
```

MOS range: 1–28 households per PSU. Total: 20,940

STATA: Step 3 — Examine MOS Distribution

STATA Code

```
. summarize n_households, detail
```

```
          n_households
-------------------------------------------
    Mean        6.507
    Std. Dev.    2.847
    Min          1
    Max          28
    Sum        20940
```

R: Step 4 — Set Sampling Parameters

R Code

```
# Define sample size
n_sample <- 100

# Total MOS
total_mos <- sum(psu_frame$n_households)

# Sampling interval
sampling_interval <- total_mos / n_sample

cat("Interval:", sampling_interval)
```

```
Interval: 209.4
```

Every 209.4 "household units" we select one PSU

STATA: Step 4 — Set Sampling Parameters

STATA Code

```
. * Define sample size
. local n_sample = 100

. * Total MOS
. quietly summarize n_households
. local total_mos = r(sum)

. * Sampling interval
. local interval = `total_mos' / `n_sample'

. display "Interval: " `interval'
```

```
Interval: 209.4
```

R: Step 5 — Selection Probabilities

R Code

```
# Calculate selection probability
psu_frame <- psu_frame %>%
mutate(
prob = n_sample * n_households / total_mos
)

# Check range
range(psu_frame$prob)
```

```
[1] 0.004776 0.133715
```

$\pi_{min} = 0.48\%$ (MOS=1), $\pi_{max} = 13.4\%$ (MOS=28)

All probabilities < 1 — no certainty units!

R: Step 6 — Check for Certainty Units

R Code

```
# Flag certainty units (prob > 1)
psu_frame <- psu_frame %>%
mutate(certainty = prob > 1)

# Count certainty units
sum(psu_frame$certainty)
```

```
[1] 0
```

✔ **No certainty units** — all PSUs can be selected via PPS

We'll discuss what to do when certainty units exist shortly...

R: Step 7 — Randomize & Cumulative MOS

R Code

```
# Set seed for reproducibility
set.seed(42)

# Randomize order (RPPS)
psu_frame <- psu_frame %>%
mutate(rand = runif(n())) %>%
arrange(rand) %>%
mutate(seq_num = row_number())

# Calculate cumulative MOS
psu_frame <- psu_frame %>%
mutate(
cum_mos = cumsum(n_households),
cum_mos_prev = lag(cum_mos, default = 0)
)
```

R: Cumulative MOS Structure

R Code

```
head(psu_frame[, c("seq_num", "n_households",
"cum_mos_prev", "cum_mos")], 6)
```

```
  seq_num n_households cum_mos_prev cum_mos
1       1           7           0       7
2       2           6           7      13
3       3           5          13      18
4       4           8          18      26
5       5           4          26      30
6       6           6          30      36
```

PSU 1 spans $(0, 7]$, PSU 2 spans $(7, 13]$, etc.

R: Step 8 — Random Start

R Code

```r
# Random start in [0, interval)
random_start <- runif(1, min = 0, max = sampling_interval)

cat("Random start:", round(random_start, 2))
```

```
Random start: 87.32
```

```r
# Generate all selection points
selection_points <- random_start +
(0:(n_sample - 1)) * sampling_interval

head(selection_points, 5)
```

```
[1]  87.32  296.72  506.12  715.52  924.92
```

R: Step 9 — Selection Algorithm

R Code

```
# Initialize selection flag
psu_frame$selected <- FALSE

# Match each point to a PSU
for (point in selection_points) {
idx <- which(
psu_frame$cum_mos_prev < point &
psu_frame$cum_mos >= point
)
if (length(idx) > 0) {
psu_frame$selected[idx[1]] <- TRUE
}
}

sum(psu_frame$selected)
```

```
[1] 100
```

R: Step 10 — Extract the Sample

R Code

```
# Extract selected PSUs
pps_sample <- psu_frame %>%
filter(selected == TRUE) %>%
select(psu, stratum, prov, n_households, prob)

# Verify sample
nrow(pps_sample)
mean(pps_sample$n_households)
```

```
[1] 100
[1] 7.1
```

Sample mean MOS (7.1) > Frame mean (6.5) — PPS confirmed!

R: Step 11 — Base Weights

R Code

```
# Base weight = 1 / selection probability
pps_sample <- pps_sample %>%
mutate(base_weight = 1 / prob)

# Verify: sum should equal frame size
sum(pps_sample$base_weight)
```

```
[1] 3218
```

✔ **Weight verification passed!**

$\sum w_i = 3{,}218 = N$ (frame size)

STATA: Complete Selection Code

STATA Code

```
. * Set seed and randomize
. set seed 42
. gen rand = runiform()
. sort rand
. * Cumulative MOS
. gen cum_mos = sum(n_households)
. gen cum_mos_prev = cum_mos - n_households
. * Random start and selection
. local interval = 20940 / 100
. local start = runiform() * `interval'
. gen selected = 0
. forvalues k = 0/99 {
.    local point = `start' + `k' * `interval'
.    quietly replace selected = 1 ///
if cum_mos_prev < `point' & cum_mos >= `point' & selected == 0
. }
```

STATA: Verify Selection

STATA Code

```
. * Count selected
. count if selected == 1
```

```
  100
```

```
. * Mean MOS in sample
. summarize n_households if selected == 1
```

```
    Variable |        Obs        Mean
-------------+---------------------------------------
n_households |        100         7.1
```

Certainty Units

When PSUs Are Too Big to Skip

What Are Certainty Units?

Definition: PSUs with selection probability $\pi_i > 1$

When does this happen?

$$\pi_i = \frac{n \times \text{MOS}_i}{\sum \text{MOS}} > 1$$

$$\Rightarrow \text{MOS}_i > \frac{\sum \text{MOS}}{n} = I$$

In words: A PSU's size exceeds the sampling interval

Example: If $I = 209.4$ and a PSU has MOS $= 300$
$\rightarrow \pi = 100 \times 300/20940 = 1.43 > 1$

Why Is This a Problem?

Problems:

- Multiple selection points land in same PSU
- PSU would be selected multiple times
- Probability > 1 is mathematically invalid
- Weights would be negative (impossible!)

Handling Certainty Units: The Solution

Two-phase approach:

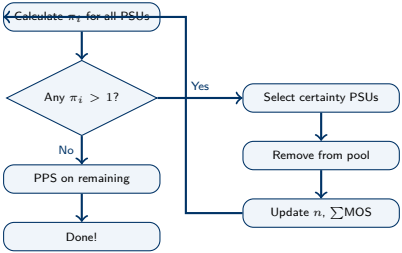
Phase 1: Identify & Remove Certainty Units

1. Calculate π_i for all PSUs
2. Flag PSUs where $\pi_i > 1$
3. Select these with **certainty** ($\pi = 1$)
4. Remove from PPS pool

Phase 2: PPS for Remaining PSUs

1. Reduce sample size: $n' = n - n_{certainty}$
2. Reduce total MOS: $\sum \text{MOS}' = \sum \text{MOS} - \sum \text{MOS}_{certainty}$
3. Recalculate interval: $I' = \sum \text{MOS}'/n'$
4. Check again for new certainty units
5. Repeat until no certainty units remain

Certainty Unit Algorithm



R: Handling Certainty Units

R Code

```
handle_certainty <- function(frame, n) {
total_mos <- sum(frame$mos)
frame$prob <- n * frame$mos / total_mos

# Identify certainty units
certainty <- frame %>% filter(prob > 1)

if (nrow(certainty) > 0) {
# Select with certainty
certainty$prob <- 1
certainty$weight <- 1

# Remove and recurse
remaining <- frame %>% filter(prob <= 1)
n_new <- n - nrow(certainty)

result <- handle_certainty(remaining, n_new)
return(bind_rows(certainty, result))
}
return(frame)
}
```

STATA: Handling Certainty Units

STATA Code

```
* Loop until no certainty units
local done = 0
local n_remaining = `n_sample'
while `done' == 0 {
* Calculate probabilities
egen temp_mos = sum(n_households) if certainty != 1
gen prob = `n_remaining' * n_households / temp_mos
* Check for certainty
count if prob > 1 & certainty != 1
if r(N) == 0 {
local done = 1
}
else {
replace certainty = 1 if prob > 1
count if certainty == 1
local n_remaining = `n_sample' - r(N)
}
drop temp_mos prob
}
```

Zambara: No Certainty Units

Check for Zambara GHS:

Parameter	Value
Sampling interval (I)	209.4
Maximum PSU size	28
Max selection probability	0.134

$$\text{Max MOS} = 28 < 209.4 = I$$

✔ **No PSU is large enough to be a certainty unit**

A PSU would need MOS > 209 to be a certainty unit.
The largest PSU has only 28 households.

When Do Certainty Units Occur?

Common scenarios:

1. **Highly skewed populations**
   Few very large clusters (cities) among many small ones
2. **Small sample sizes**
   Smaller n means larger interval, but if one PSU dominates...
3. **Establishment surveys**
   Large companies vs. small businesses
4. **Hospital/school surveys**
   Major hospitals vs. small clinics

Zambara example where certainty would occur:

If we only wanted $n = 10$ PSUs:
$I = 20940/10 = 2094$. Still no certainty (max MOS = 28).

But if one PSU had 3,000 households:
$\pi = 10 \times 3000/23940 = 1.25 > 1 \rightarrow$ Certainty!

Coding the Selection: Summary

Key Implementation Steps:

1. Load data and create PSU-level frame with MOS
2. Set n, calculate $\sum$MOS, compute interval I
3. Calculate selection probabilities π_i
4. Check for certainty units ($\pi > 1$)
5. Randomize PSU order (RPPS)
6. Calculate cumulative MOS
7. Generate random start $r \in [0, I)$
8. Create selection points: $r, r + I, r + 2I, \ldots$
9. Match points to PSUs
10. Calculate base weights: $w_i = 1/\pi_i$
11. Verify: $\sum w_i \approx N$

Next: Module 2.3 — STATA `svy` Commands

Module 2.3

GIS Fundamentals for Statisticians

Spatial Data Integration for Survey Design

Slides 121–150

Why GIS for Survey Statisticians?

Zambara Case

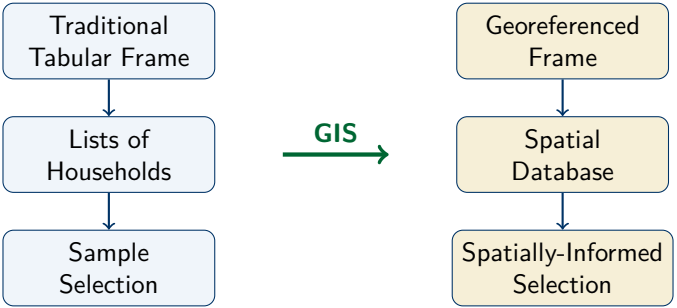
Lindiwe realizes her sampling frame lacks geographic coordinates. How can she ensure spatial coverage?

GIS transforms survey methodology through:

- **Frame Enhancement** — Link boundaries to sampling units
- **Spatial Stratification** — Stratify by geography
- **Coverage Assessment** — Visualize sample gaps
- **Field Operations** — Optimize routes and workloads

Key: Maps reveal patterns invisible in tables — "adequate" provincial samples may have zero coverage in remote districts.

The Spatial Revolution in Survey Methodology



Benefits: Satellite imagery for frame updates, GPS field verification, spatial sampling algorithms, real-time dashboards.

Source: Adapted from Grosh & Muñoz (1996), World Bank LSMS

Key GIS Concepts: Data Types

Vector Data

- **Points** — PSU centroids, dwellings
- **Lines** — Roads, rivers
- **Polygons** — Districts, EAs

Raster Data

- Satellite imagery
- Population density grids
- Land use classifications

Attribute Data

- Tabular info linked to geometry
- Population counts by district
- Survey estimates by region

Spatial Operations

- Overlay (point-in-polygon)
- Buffer zones, spatial joins

Source: Longley et al. (2015), Geographic Information Systems and Science

Coordinate Reference Systems (CRS)

A CRS defines how coordinates map to Earth's surface.

Geographic CRS

- Latitude/longitude (degrees)
- Example: WGS 84 (EPSG:4326)
- GPS devices use this

Projected CRS

- Flat 2D (meters)
- Example: UTM Zone 35S
- Required for distance/area

⚠ **Warning:** Mixing CRS causes misalignment! Always verify data sources use the same CRS before spatial joins.

EPSG = European Petroleum Survey Group geodetic registry

CRS for Southern Africa

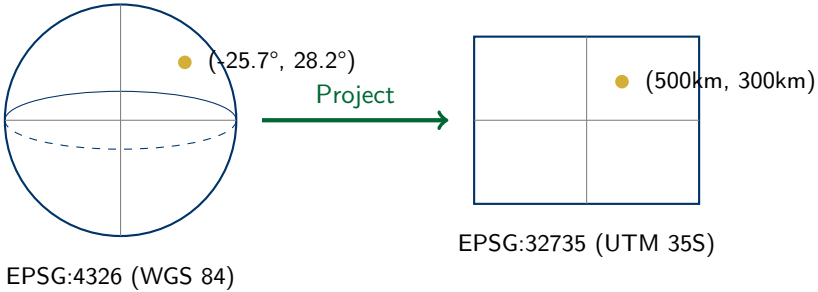
CRS Name	EPSG	Best Use
WGS 84	4326	GPS data, web maps
Hartebeesthoek94	4148	South Africa official
UTM Zone 35S	32735	Central SA (Gauteng)
Africa Albers	102022	Continental comparisons

R Code to Check/Set CRS:

```
st_crs(shapefile)            # Check CRS
st_transform(shp, 4326)   # Reproject
```

Source: EPSG Geodetic Parameter Registry, https://epsg.io

Understanding EPSG Codes



Key: EPSG:4326 = degrees; EPSG:32735 = meters. Use projected CRS for distance/area calculations.

CRS in Practice: R and STATA

R with sf

```r
library(sf)
# Load shapefile
districts <- st_read(
  "gadm41_ZAF_2.shp")
# Check CRS
st_crs(districts)
# Reproject for area calc
districts_utm <- st_transform(
  districts, crs = 32735)
# Calculate area (sq km)
districts_utm$area_km2 <-
  st_area(districts_utm) / 1e6
```

STATA with shp2dta

```stata
* Convert shapefile
shp2dta using "gadm41_ZAF_2", ///
  database(zaf_db) ///
  coordinates(zaf_shp) ///
  genid(id) replace

* Note: shp2dta reads
* projection from .prj file

* For reprojection:
ssc install geo2xy
geo2xy _Y _X, proj(web_mercator)
```

Source: Lumley (2010); StataCorp Spatial Data Reference

Anatomy of a Shapefile

A shapefile is a **collection of files** storing vector data:

Extension	Required	Contents
`.shp`	✓	Geometry (polygons)
`.shx`	✓	Shape index
`.dbf`	✓	Attribute table
`.prj`	Recommended	CRS definition
`.cpg`	Optional	Character encoding

⚠ **Warning:** Always keep all shapefile components together! Missing `.prj` causes CRS confusion.

Shapefile vs Modern Formats

Shapefile

- ✔ Universal support
- ✔ Simple structure
- ✘ Multiple files
- ✘ 2GB limit

GeoPackage

- ✔ Single file
- ✔ No size limit
- ✔ Multiple layers
- ⚠ Less legacy support

GeoJSON

- ✔ Human-readable
- ✔ Web-friendly
- ✘ Large files
- ✘ Slow for big data

For SADC NSOs: Shapefile remains the *lingua franca*. Consider GeoPackage for internal workflows.

GADM: Global Administrative Boundaries

GADM (gadm.org) provides free administrative boundary data:

Level	Description	South Africa	Count
0	Country	South Africa	1
1	Province	Western Cape, ...	9
2	District	Cape Town, eThekwini, ...	52
3	Local Municipality	Stellenbosch, ...	213

Workshop Files:

- `gadm41_ZAF_2.shp` — 52 district polygons
- Attributes: `NAME_1` (province), `NAME_2` (district)

Source: GADM database v4.1

OCHA: Humanitarian Data Exchange

OCHA (data.humdata.org) provides UN-standardized boundaries.

Advantages over GADM:

- UN-endorsed naming
- P-codes (unique IDs)
- Regular updates
- Humanitarian standard

P-Code Example:

```
ZAF --- Country
ZAF-EC --- E. Cape
ZAF-EC-BUF --- Buffalo City
```

Source: UN OCHA Centre for Humanitarian Data

Loading Shapefiles: R vs STATA

R with sf

```r
library(sf)
# Load GADM districts
gadm <- st_read(
  "gadm41_ZAF_shp/gadm41_ZAF_2.shp")

# Explore
print(gadm)
# 52 features, 13 fields
# Geometry: MULTIPOLYGON
# CRS: WGS 84

names(gadm)
# "GID_2" "NAME_1" "NAME_2"...
```

STATA with shp2dta

```stata
* Install packages
ssc install shp2dta
ssc install spmap

* Convert shapefile
shp2dta using ///
  "gadm41_ZAF_shp/gadm41_ZAF_2",///
  database(gadm_db) ///
  coordinates(gadm_shp) ///
  genid(district_id) replace

use gadm_db, clear
describe
```

Note: STATA requires conversion; R reads shapefiles directly.

Spatial DataFrames: The sf Object

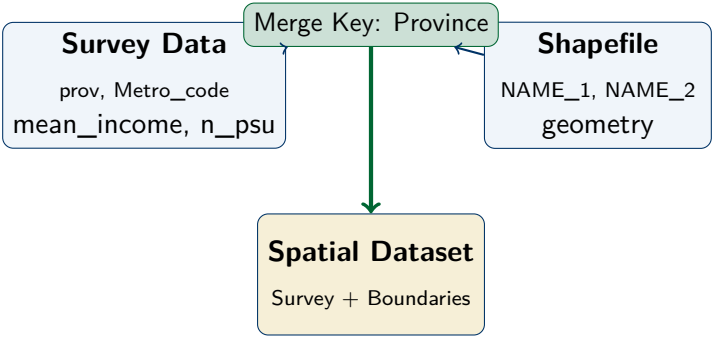
The sf package creates "simple feature" data frames:

```
> print(gadm_districts, n=3)
Simple feature collection with 52 features and 5 fields
Geometry type: MULTIPOLYGON | CRS: WGS 84
GID_2         NAME_1        NAME_2          TYPE_2
1 ZAF.1.1_1  Western Cape  City of Cape Town  Metro
2 ZAF.1.2_1  Western Cape  West Coast        District
3 ZAF.1.3_1  Western Cape  Cape Winelands    District
```

Key Features:

- Works like data.frame with dplyr verbs
- Geometry column stores spatial info
- CRS embedded in object

Integrating Survey Data with Geography



Challenge: Survey uses codes (prov=1–9); shapefile uses names. **Solution:** Create crosswalk table.

Merging Survey Data to Shapefile

R Approach

```r
# Provincial estimates
prov_est <- ghs_hh %>%
  group_by(prov_name) %>%
  summarise(
    mean_income = weighted.mean(
      totmhinc, house_wgt),
    n_psu = n_distinct(psu))

# Dissolve to provinces
gadm_prov <- gadm_districts %>%
  group_by(NAME_1) %>%
  summarise(geom = st_union(geom))

# Merge
prov_spatial <- gadm_prov %>%
  left_join(prov_est,
    by = c("NAME_1" = "prov_name"))
```

STATA Approach

```stata
* Create estimates
use ghs_hh, clear
collapse (mean) mean_income ///
  (count) n_psu = psu ///
  [pw=house_wgt], by(prov)

* Add province names
gen prov_name = ""
replace prov_name = ///
  "Western Cape" if prov==1
// ... repeat for 2-9

* Merge to shapefile
use gadm_db, clear
merge m:1 prov_name using ///
  prov_estimates, nogen
```

Zambara: Survey-Shapefile Integration Results

Zambara Case

Lindiwe merged GHS 2024 with GADM boundaries. Now she can visualize estimates geographically.

Province	Districts	PSUs	HH	Mean Inc. (R)
Western Cape	2	338	1,871	23,323
Eastern Cape	3	430	2,666	8,889
Northern Cape	1	141	840	13,123
KwaZulu-Natal	2	504	3,153	11,551
Gauteng	4	837	5,413	15,622
(4 more provinces)				
Total	**17**	**3,218**	**20,940**	—

Source: Lab 2.3 output, GHS 2024 + GADM 4.1

Creating Choropleth Maps

Choropleth: Areas shaded proportional to a variable.

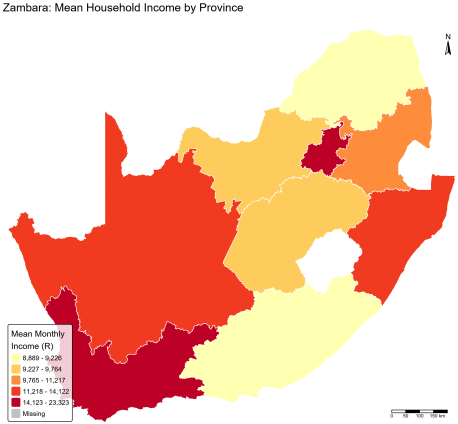
R with tmap

```
library(tmap)
tm_shape(prov_spatial) +
  tm_polygons(
    col = "mean_income",
    palette = "YlOrRd",
    style = "quantile",
    n = 5,
    title = "Mean Income (R)"
  ) +
  tm_layout(
    title = "Zambara: Income")
```

STATA with spmap

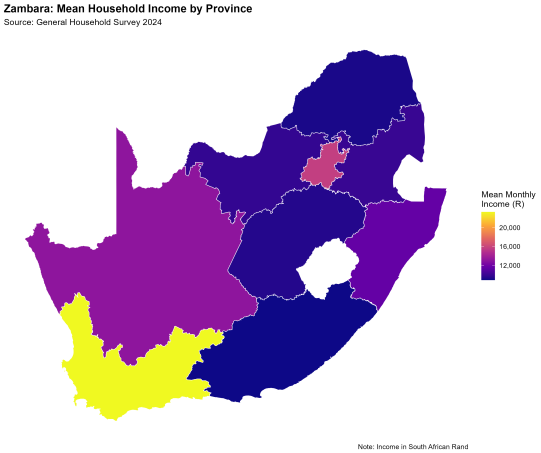
```
use gadm_merged, clear

spmap mean_income ///
  using gadm_shp, ///
  id(district_id) ///
  clmethod(quantile) ///
  clnumber(5) ///
  fcolor(YlOrRd) ///
  title("Zambara: Income")

graph export ///
  "income_map.png", replace
```

Zambara: Mean Household Income by Province



Source: Lab 2.3 — GHS 2024 + GADM, tmap visualization

Alternative Visualization: ggplot2 + viridis



Source: Lab 2.3 — viridis "plasma" palette (colorblind accessible)

Spatial Stratification for Sampling

Spatial Stratification: Using geographic boundaries to create sampling strata.

Statistical Benefits:

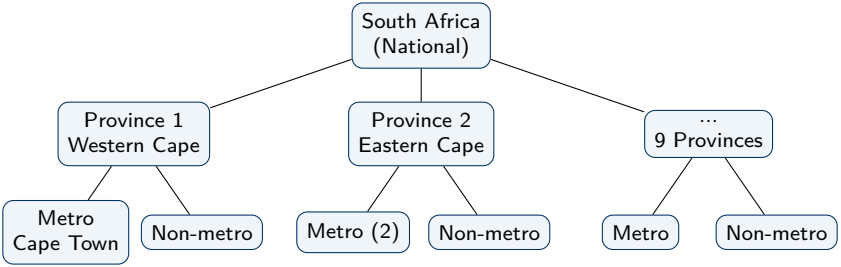
- Reduced variance (spatial autocorrelation)
- Guaranteed minimum per region
- Implicit geographic proportionality

Operational Benefits:

- Organized fieldwork by region
- Local supervisors per stratum
- Easier logistics

GHS 2024: 248 strata from Province $\times$ Metro $\times$ GeoType (Urban/Traditional/Farms).

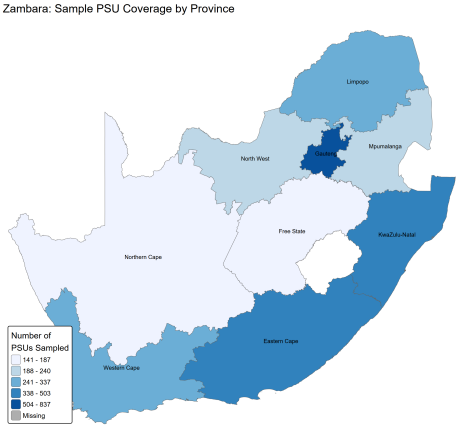
GHS 2024: Spatial Stratification Structure



Further stratified by GeoType: Urban, Traditional, Farms

Source: GHS 2024 Metadata, Statistics South Africa

Zambara: Sample PSU Coverage by Province



Source: Lab 2.3 — PSUs range from 141 (Northern Cape) to 837 (Gauteng)

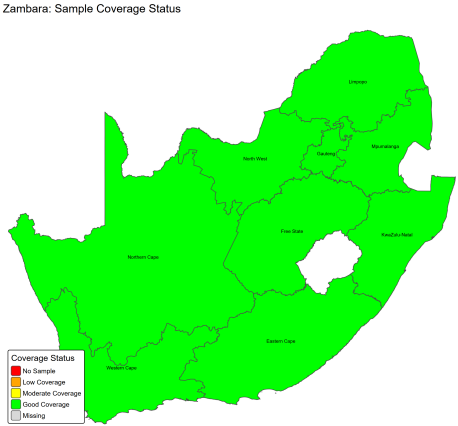
Identifying Spatial Gaps in Coverage

Key Question: Are any areas systematically under-sampled?

Metric	Value	Threshold	Status
Total PSUs	3,218	—	—
Metro codes	17/17	17	✔ Complete
Provinces	9/9	9	✔ Complete
Min PSUs/Province	141	$\geq$50	✔ Adequate

Finding: All provinces have adequate coverage. No spatial gaps at administrative level.

Zambara: Sample Coverage Status



Source: Lab 2.3 — Green = Good Coverage ($\geq$100 PSUs)

When Spatial Gaps Exist: Remediation

Zambara Case

In earlier surveys, Lindiwe found Northern Bushveld had zero sample in 3 of 5 districts.

Remediation Options:

1. **Increase Sample** — Add PSUs in under-covered areas
2. **Model-Based** — Small area estimation techniques
3. **Borrow Strength** — Data from similar neighbors
4. **Admin Data** — Supplement with registers
5. **Document** — Acknowledge gaps in metadata

⚠ Gaps often correlate with hard-to-reach populations: remote areas, conflict zones, informal settlements.

Advanced: Spatially Balanced Sampling

Beyond stratification: Algorithms optimizing spatial spread.

GRTS

Generalized Random Tessellation Stratified

- Spatially balanced probability sample
- Hierarchical randomization
- R: `spsurvey`

LPM

Local Pivotal Method

- Maximizes spatial spread
- Good for environmental surveys
- R: `BalancedSampling`

Application: When frame has coordinates (GPS dwelling units, EA centroids).

Source: Stevens & Olsen (2004), JASA

Lab 2.3: What We Accomplished

Spatial Frame Integration Workflow:

1. Loaded Shapefiles — GADM (52 districts), OCHA
2. Loaded Survey — GHS 2024 (20,940 HH, 3,218 PSUs)
3. Created Merge Keys — Province names, Metro_code
4. Calculated Estimates — Mean income by district
5. Merged to Geometry — Survey → polygons
6. Created Maps — Income, PSU coverage, gaps

Outputs: `lab2_3_provincial_income_map.png`, `lab2_3_psu_coverage_map.png`, `lab2_3_coverage_gaps_map.png`

Module 2.3: Key Takeaways

1. **CRS Matters** — Verify before spatial operations
2. **Shapefiles = Multiple Files** — Keep .shp/.shx/.dbf/.prj together
3. **Merge Keys Critical** — Survey codes must match shapefile names
4. **Maps Reveal Patterns** — Inequality, gaps visible at a glance
5. **R vs STATA** — Both capable; R more flexible for GIS
6. **Spatial Stratification** — Boundaries improve efficiency

GIS transforms survey methodology from tabular statistics to spatial intelligence.

Next: Module 2.4 — AI/ML Applications in Sampling

Zambara Case

With GIS mastered, Lindiwe wonders: "Can machine learning predict non-response before fieldwork?"

Coming Up:

- ML for non-response prediction
- Satellite imagery for frame construction
- AI-assisted frame updates
- Ethical considerations

Questions on GIS Fundamentals?

Break: 15 minutes

Case Study

The Empty Quarter

Discovering Spatial Gaps in Survey Coverage

Module 2.3 Continued — Slides 151–180

Case Study: Monday Morning at ZNSO

Zambara Case

Lindiwe arrives at ZNSO headquarters with coffee in hand and a nagging worry. The Minister's words echo in her mind: "Previous surveys faced criticism for urban bias..."

Her Mission Today:

- Verify GHS 2024 provides adequate geographic coverage
- Map the sample distribution across all provinces
- Identify any "blind spots" before the final report

Her Tools:

- GHS 2024 household data (20,940 observations)
- GADM administrative boundaries (shapefile)
- R with sf, tmap, and dplyr

The Question That Keeps Her Awake

"Where are we NOT collecting data?"

Are there geographic blind spots in our sample?

Why This Matters:

- Unsampled areas = zero direct information
- National estimates may be biased
- Policy decisions could overlook entire communities

Spatial gaps are invisible in summary tables but obvious on a map.

Step 1: Load and Examine the Data

Zambara Case

Lindiwe opens RStudio and loads the GHS 2024 data...

```
library(haven); library(dplyr); library(sf); library(tmap)

ghs_hh <- read_dta("ghs-2024-hhold-v1.dta")
cat("Total households:", nrow(ghs_hh), "\n")
cat("Total PSUs:", n_distinct(ghs_hh$psu), "\n")
cat("Provinces covered:", n_distinct(ghs_hh$prov), "\n")
```

Initial Output:

```
Total households: 20,940
Total PSUs: 3,218
Provinces covered: 9
```

Step 2: Province-Level Summary

Province	PSUs	HH	%	Status
Gauteng	837	5,413	25.8%	✓
KwaZulu-Natal	504	3,153	15.1%	✓
Eastern Cape	430	2,666	12.7%	✓
Western Cape	338	1,871	8.9%	✓
Limpopo	295	2,361	11.3%	✓
Mpumalanga	241	1,647	7.9%	✓
North West	234	1,317	6.3%	✓
Free State	188	1,239	5.9%	✓
Northern Cape	141	840	4.0%	✓

Zambara Case

"All provinces have sample. But wait — I need to check *within* provinces. The devil is in the details..."

Step 3: Drilling Down to Districts

Zambara Case

Lindiwe examines the Metro_code variable — 17 distinct metro/district groupings...

The Question: Are all 17 Metro_code values present?

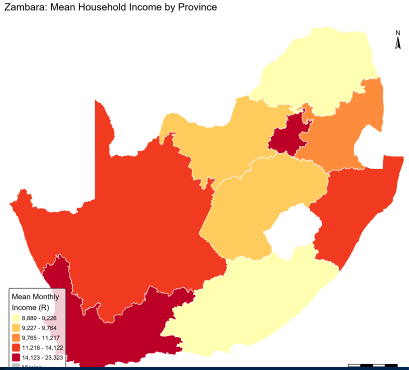
Code	Area	PSUs	HH
1	WC Non-metro	119	646
2	City of Cape Town	219	1,225
3	EC Non-metro	296	1,848
4	Buffalo City	54	368
...	...	...	...
17	Limpopo Non-metro	295	2,361

"17 codes expected, 17 found. Still no gaps at this level..."

Step 4: Mapping Income Distribution

Zambara Case

"Let me visualize the data geographically. Numbers can hide patterns that maps reveal."



Interpreting the Income Map

Key Observations:

High Income (R14,000+):

- Western Cape: R23,323
- Gauteng: R15,622
- Northern Cape: R13,123

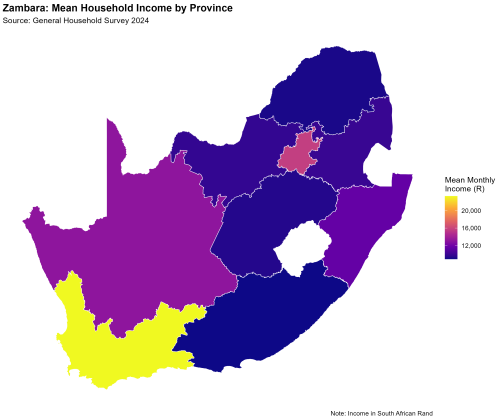
Low Income (<R10,000):

- Eastern Cape: R8,889
- Limpopo: R9,101
- Free State: R9,311

Zambara Case

"The inequality is stark. But something else catches my eye — look at the Northern Cape. It's *huge* but has the fewest PSUs..."

Alternative: Colorblind-Accessible Palette

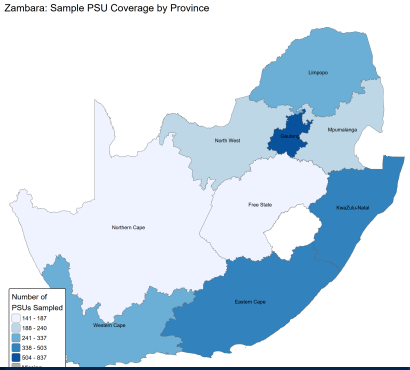


Source: Lab 2.3 — viridis "plasma" palette (colorblind accessible)

Step 5: Mapping PSU Distribution

Zambara Case

"Let me map the actual sample distribution. Where exactly are our 3,218 PSUs located?"



Step 6: The Discovery

⚠ WAIT...

Northern Cape: 141 PSUs across 372,889 km^2

That's **1 PSU per 2,644 km^2**

Zambara Case

Lindiwe's eyes widen. "One PSU covering an area larger than Luxembourg? There must be vast stretches with no sample at all!"

Comparison: Gauteng has 837 PSUs across 18,178 km^2 = 1 PSU per 22 km^2

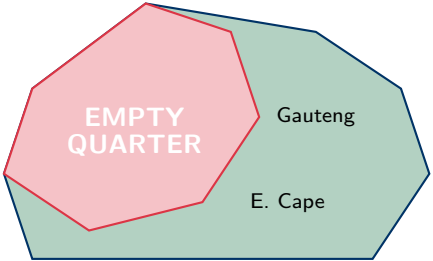
Step 7: Investigating the Empty Quarter

```
# Get Northern Cape district breakdown
nc_districts <- ghs_hh %>%
  filter(prov == 3) %>%  # Northern Cape
  group_by(district_name) %>%
  summarise(
    n_psu = n_distinct(psu),
    n_hh = n(),
    area_km2 = first(district_area)
  ) %>%
  mutate(psu_density = n_psu / area_km2 * 1000)

print(nc_districts)
```

Output:

```
Warning: 3 districts have fewer than 30 PSUs
- Namakwa: 18 PSUs across 126,836 km²
- Pixley ka Seme: 22 PSUs across 103,410 km²
- ZF Mgcawu: 24 PSUs across 102,525 km²
```

The "Empty Quarter" Revealed



Dense Coverage

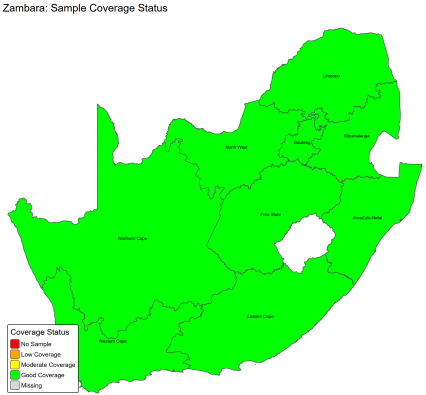
Sparse Coverage

Northern Cape:
30% of land area
4% of sample

Zambara Case

"I've found it — the Empty Quarter. Three districts covering 332,771 km^2 with only 64 PSUs between them."

Step 8: Coverage Status Assessment



Source: Lab 2.3 — Coverage assessment by province

Quantifying the Coverage Gap

District	Area (km²)	PSUs	km²/PSU	Status
City of Cape Town	2,461	219	11	✔ Dense
eThekwini	2,556	185	14	✔ Dense
City of Johannesburg	1,645	287	6	✔ Dense
Namakwa	126,836	18	7,046	✘ Sparse
Pixley ka Seme	103,410	22	4,700	✘ Sparse
ZF Mgcawu	102,525	24	4,272	✘ Sparse

⚠ The Empty Quarter districts have 300–600x lower PSU density than metro areas.

Why Do Empty Quarters Exist?

Common Causes of Spatial Gaps:

Structural Factors:

- Low population density
- Probability proportional to size (PPS) selection
- Budget constraints for remote areas
- Frame based on populated EAs only

Operational Factors:

- Inaccessibility (roads, weather)
- Security concerns
- High travel costs per interview
- Interviewer availability

Zambara Case

"The Northern Cape has 2% of South Africa's population spread across 30% of its land. PPS sampling naturally produces sparse coverage here."

Is the Empty Quarter a Problem?

It Depends on the Survey Objective:

Not a Problem If:

- Goal is national estimates only
- Provincial estimates acceptable
- Population is truly sparse
- Domain estimation not required

A Problem If:

- District-level estimates needed
- Sparse populations are policy targets
- Geographic equity is required
- Mining/farming communities matter

The Northern Cape's mining and agricultural communities may have unique characteristics invisible to the survey.

Lindiwe's Dilemma

Zambara Case

Lindiwe faces a difficult choice. The Minister wants "national representativeness" but what does that mean for the Empty Quarter?

Option A: Accept the Design

- PPS is statistically valid
- Provincial estimates are reliable
- Document limitations transparently

Option B: Recommend Changes

- Oversample sparse regions
- Add supplemental PSUs
- Increase future survey budget

Remediation Strategy 1: Supplemental Sampling

Add PSUs to Under-Covered Areas:

1. Identify districts below threshold (e.g., <30 PSUs)
2. Select additional PSUs from frame
3. Deploy field teams to new locations
4. Calculate separate weights for supplement

Cost-Benefit:

Pros	Cons
Direct measurement	High cost per interview
Reduces bias	Extended timeline
Complete coverage	Complex weighting

⚠ Northern Cape interviews cost 3x more than Gauteng due to travel distances.

Remediation Strategy 2: Small Area Estimation

Model-Based Estimates for Sparse Areas:

Small Area Estimation (SAE) borrows strength from:

- Similar neighboring areas with adequate sample
- Auxiliary data (census, admin records, satellite)
- Covariates correlated with outcomes

Basic Model:

$$\hat{Y}_d = \mathbf{X}_d \hat{\boldsymbol{\beta}} + \hat{u}_d$$

Where $\hat{u}_d$ is a random effect for district d.

SAE can produce district estimates even with 18 PSUs, but confidence intervals will be wide.

Remediation Strategy 3: Administrative Data

Supplement Survey Gaps with Records:

Northern Cape Sources:

- Mining company employment records
- Agricultural census data
- Social grant registries
- Municipal service records
- School enrollment

Challenges:

- Coverage bias
- Definition mismatches
- Data quality varies
- Access restrictions

Admin data works best as complement, not replacement. Use for triangulation.

Remediation Strategy 4: Transparent Documentation

If gaps cannot be fixed, document clearly:

Required in Methodology Report:

1. List of districts with sparse coverage
2. Reasons (low population density, PPS design)
3. Population affected (count and %)
4. Impact on estimates
5. Recommendations for users

Example Statement:

> "Three Northern Cape districts (Namakwa, Pixley ka Seme, ZF Mgcawu) have fewer than 25 PSUs each due to low population density. District-level estimates for these areas have coefficients of variation exceeding 25% and should be interpreted with caution."

Lindiwe's Decision

Zambara Case

After careful analysis, Lindiwe prepares her recommendation for the Director...

Her Recommendation:

1. **Accept Current Design** for GHS 2024
   - National and provincial estimates are valid
   - Redesign mid-survey is not feasible
2. **Document Limitations** transparently
   - Flag Empty Quarter in metadata
   - Publish CVs for all domain estimates
3. **Recommend Future Changes**
   - Oversample sparse regions in GHS 2025
   - Explore SAE for district estimates

Quality Metrics for Coverage Assessment

Standard Metrics to Report:

Metric	GHS 2024	Target
Province Coverage	9/9 (100%)	100%
Metro Area Coverage	17/17 (100%)	100%
Min PSUs per Province	141	$\geq$100
Min PSUs per District	18	$\geq$30
Districts Below Threshold	3	0
Population in Sparse Areas	2.1%	$<$5%

GHS 2024 meets all targets except district-level minimum PSUs in the Empty Quarter.

Presenting to the Director

Zambara Case

Tuesday afternoon. Lindiwe presents her findings to Director Molefe...

"Director, I have good news and a concern."

Good News:

- All 9 provinces adequately covered
- All 17 metro areas have sample
- National estimates will be reliable

The Concern:

- Three Northern Cape districts are sparse
- District-level estimates unreliable there
- We should document this limitation

The Director's Response

Zambara Case

Director Molefe nods slowly. "You've done excellent work, Lindiwe. This is exactly the kind of quality assurance we need."

Director's Decisions:

1. Approve GHS 2024 methodology documentation with coverage caveats
2. Request budget increase for GHS 2025 to oversample Northern Cape
3. Commission feasibility study for small area estimation
4. Present findings to Statistics Council

Transparency about limitations builds trust. Hiding problems destroys credibility.

Lessons for SADC National Statistics Offices

Every Country Has an "Empty Quarter":

Common Examples:

- Botswana: Kalahari Desert
- Namibia: Skeleton Coast
- Zambia: Remote Western Province
- Mozambique: Conflict-affected Cabo Delgado
- DRC: Inaccessible forest regions

Universal Challenges:

- Sparse populations
- Vast distances
- Limited infrastructure
- Security concerns
- Budget constraints

⚠ The populations in Empty Quarters are often the most vulnerable and most in need of policy attention.

Coverage Assessment Workflow

Replicable Steps for Any Survey:

1. **Load Data** — Survey + administrative shapefile
2. **Calculate Metrics** — PSUs per admin unit, area per PSU
3. **Set Thresholds** — Define "adequate" coverage
4. **Map Distribution** — Choropleth of PSU counts
5. **Identify Gaps** — Flag units below threshold
6. **Assess Impact** — Population affected, CV estimates
7. **Document** — Include in methodology report

This workflow takes 2–3 hours and should be standard practice for every survey.

Case Study: Key Takeaways

1. **Provincial OK $\neq$ District OK** — Always drill down
2. **Maps Reveal Hidden Patterns** — Tables can mask spatial gaps
3. **PPS Creates Sparse Regions** — By design, not by accident
4. **Document Limitations** — Transparency builds trust
5. **Plan Remediation** — SAE, oversampling, admin data
6. **Budget for Coverage** — Remote areas cost more

Zambara Case

Lindiwe learned that finding the Empty Quarter is not failure — ignoring it is.

Next: Module 2.4 — AI/ML in Sampling

Zambara Case

Coverage mapped, Lindiwe now wonders: "Can machine learning help predict which households will refuse to participate?"

Coming Up:

- ML for non-response prediction
- Satellite imagery for frame construction
- AI-assisted dwelling detection
- Ethical considerations

Questions on the Empty Quarter Case Study?

Lab Exercise: Apply this workflow to your country's survey data

Spatial Sampling Techniques

Grid-Based Methods & Frame Validation

Module 2.3 Continued — Slides 181–210

Beyond Traditional Sampling: Spatial Methods

Zambara Case

Having mapped the Empty Quarter, Lindiwe asks: "Are there better ways to ensure spatial spread in our sample?"

Traditional vs. Spatial Sampling:

Traditional (List-Based):

- Sample from enumerated frame
- No explicit spatial control
- May cluster geographically

Spatial Methods:

- Use coordinates explicitly
- Guarantee spatial spread
- Reduce spatial autocorrelation

Spatial sampling is especially valuable when outcomes are spatially correlated (poverty, health, environment).

When to Use Spatial Sampling Methods

Spatial sampling is advantageous when:

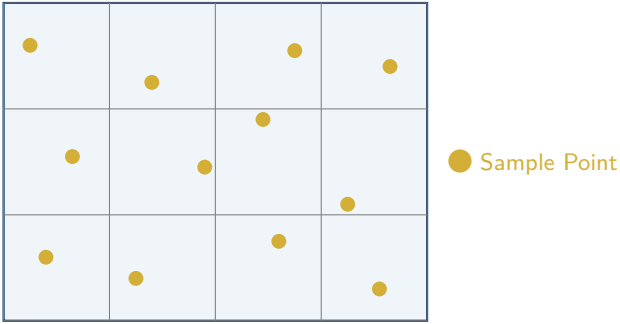
1. **Frame has coordinates** — GPS points, EA centroids
2. **Outcomes are spatially autocorrelated** — Nearby units similar
3. **Geographic coverage is priority** — Policy requires regional data
4. **Environmental/agricultural surveys** — Land, water, crops
5. **No complete list frame** — Must sample locations, then list

Examples:

- Agricultural production surveys (sample fields)
- Environmental monitoring (sample sites)
- Household surveys with georeferenced EAs

Grid-Based Sampling: The Concept

Overlay a regular grid on the study area, then sample within cells.

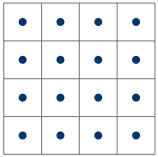


Study Area with Grid Overlay

Key Feature: Guarantees minimum spacing between sample units.

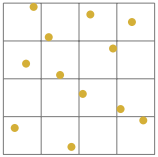
Types of Grid-Based Sampling

Systematic Grid



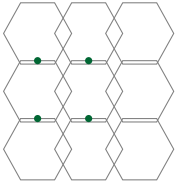
Fixed point per cell
Simple but predictable

Random within Grid



Random point per cell
Spatial spread + randomness

Hexagonal Grid



More uniform coverage
Better for circular ranges

Hexagonal grids provide more uniform neighbor distances than square grids.

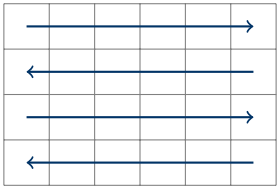
Systematic Spatial Sampling

Select every k-th unit after random start, applied spatially:

Procedure:

1. Sort frame by geographic coordinates (serpentine path)
2. Calculate sampling interval: $k = N/n$
3. Select random start $r \in [1, k]$
4. Select units at positions $r, r + k, r + 2k, \ldots$

Serpentine Ordering:



Geographic ordering ensures spatial spread

GRTS: Generalized Random Tessellation Stratified

A spatially balanced probability sampling method.

Key Properties:

- Equal probability selection (unbiased)
- Spatially balanced (spread across area)
- Hierarchically ordered (can add/remove samples)
- Applicable to points, lines, and polygons

How It Works:

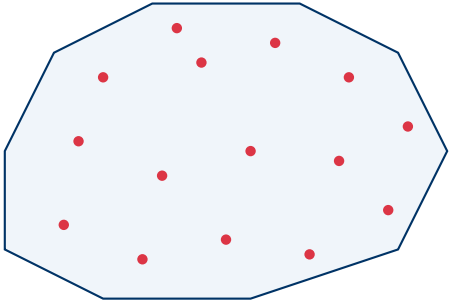
1. Map study area to unit square using hierarchical addressing
2. Apply 1-D systematic sampling on transformed coordinates
3. Back-transform to original geography

Source: Stevens & Olsen (2004), "Spatially Balanced Sampling of Natural Resources," JASA

GRTS Implementation in R

```
# Install and load spsurvey package
install.packages("spsurvey")
library(spsurvey)
library(sf)

# Load study area shapefile
study_area <- st_read("zambara_districts.shp")

# Select GRTS sample of 100 points
grts_sample <- grts(
  sframe = study_area,       # Spatial frame
  n_base = 100,              # Sample size
  seltype = "equal",         # Equal probability
  DesignID = "ZambaraSample"
)

# View selected sites
head(grts_sample$sites_base)
plot(st_geometry(study_area))
plot(st_geometry(grts_sample$sites_base), add = TRUE,
     col = "red", pch = 16)
```

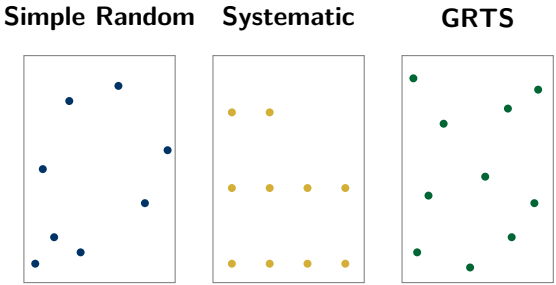
GRTS Sample: Spatial Distribution



GRTS Sample: n=15, Well-Spread

Notice: Points are spread evenly across the study area, not clustered.

Comparison: SRS vs Systematic vs GRTS



Property	SRS	Systematic	GRTS
Spatial spread	Poor	Good	Excellent
Unbiased	Yes	Yes*	Yes
Variance estimation	Easy	Difficult	Moderate
Flexible sample size	Yes	Limited	Yes

Local Pivotal Method (LPM)

Another spatially balanced sampling approach:

Algorithm:

1. Assign inclusion probabilities to all units
2. Iteratively update probabilities based on neighbors
3. Select units probabilistically, spreading selection spatially

R Implementation:

```
library(BalancedSampling)
# Coordinates matrix
coords <- cbind(frame$x, frame$y)
# Select n=100 with LPM
sample_idx <- lpm2(prob, coords)
```

LPM is particularly effective for unequal probability sampling with spatial balance.

Spatial Frame Validation: Why It Matters

Zambara Case

Lindiwe received a new sampling frame with GPS coordinates for all EAs. "How do I know these coordinates are correct?"

Common Frame Quality Issues:

- **Missing coordinates** — Units with no location
- **Invalid coordinates** — Outside country boundaries
- **Duplicate locations** — Multiple units at same point
- **Swapped lat/long** — Coordinates transposed
- **Wrong CRS** — Projection mismatch

A frame with 5% coordinate errors can produce seriously biased spatial estimates.

Validation Step 1: Check for Missing Coordinates

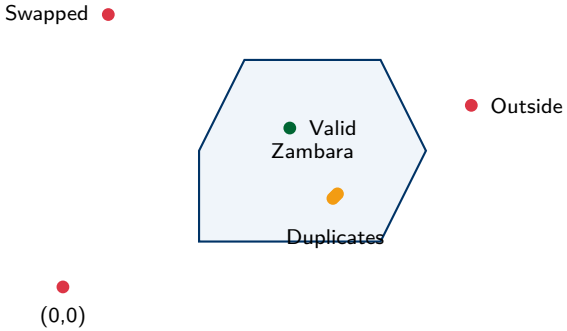
```
# Load frame with coordinates
frame <- read_dta("ea_frame_with_coords.dta")

# Check for missing coordinates
missing_lat <- sum(is.na(frame$latitude))
missing_lon <- sum(is.na(frame$longitude))

cat("Missing latitude:", missing_lat,
    "(", round(100*missing_lat/nrow(frame), 1), "%)\n")
cat("Missing longitude:", missing_lon,
    "(", round(100*missing_lon/nrow(frame), 1), "%)\n")

# Flag records with missing coordinates
frame$coord_missing <- is.na(frame$latitude) |
                       is.na(frame$longitude)

# Decision: Can these be geocoded? Imputed? Excluded?
```

Acceptable Threshold: $<2\%$ missing; $>5\%$ requires remediation.

Validation Step 2: Check Against Boundaries

```
library(sf)

# Convert frame to spatial object
frame_sf <- st_as_sf(frame,
                     coords = c("longitude", "latitude"),
                     crs = 4326)

# Load country boundary
country <- st_read("zambara_boundary.shp")

# Check which points fall outside
outside <- !st_within(frame_sf, country, sparse = FALSE)
n_outside <- sum(outside)

cat("Points outside country boundary:", n_outside, "\n")

# Map the outliers
plot(st_geometry(country))
plot(st_geometry(frame_sf[outside, ]),
     add = TRUE, col = "red", pch = 16)
```

Common Coordinate Errors



Error Types:

- Red = Invalid (outside, swapped, null island)
- Orange = Suspicious (duplicates, clusters)
- Green = Valid

Validation Step 3: Detect Duplicate Locations

```
# Round coordinates to detect near-duplicates
frame$lat_round <- round(frame$latitude, 4)
frame$lon_round <- round(frame$longitude, 4)

# Find duplicates
duplicates <- frame %>%
  group_by(lat_round, lon_round) %>%
  filter(n() > 1) %>%
  arrange(lat_round, lon_round)

cat("Duplicate coordinate pairs:",
    n_distinct(paste(duplicates$lat_round,
                     duplicates$lon_round)), "\n")
cat("Total records at duplicate locations:",
    nrow(duplicates), "\n")

# Are these legitimate? (e.g., apartments in same building)
# Or errors? (e.g., default coordinates assigned)
```

Some duplicates are valid (apartments). Mass duplicates at one point indicate data entry errors.

Validation Step 4: Detect Swapped Lat/Long

```
# South Africa expected ranges:
# Latitude: -35 to -22 (South)
# Longitude: 16 to 33 (East)

# Check for likely swaps
frame$likely_swap <- (frame$latitude > 0) |       # Should be negative
                     (frame$latitude < -40) |     # Too far south
                     (frame$longitude < 10) |     # Too far west
                     (frame$longitude > 40)       # Too far east

n_swaps <- sum(frame$likely_swap, na.rm = TRUE)
cat("Likely swapped or invalid coordinates:", n_swaps, "\n")

# Attempt automatic correction
frame$lat_corrected <- ifelse(frame$longitude < 0 &
                              frame$latitude > 0,
                              frame$longitude,  # Swap
                              frame$latitude)
```

QGIS for Visual Frame Validation

QGIS (free, open-source) provides visual validation tools:

Workflow:

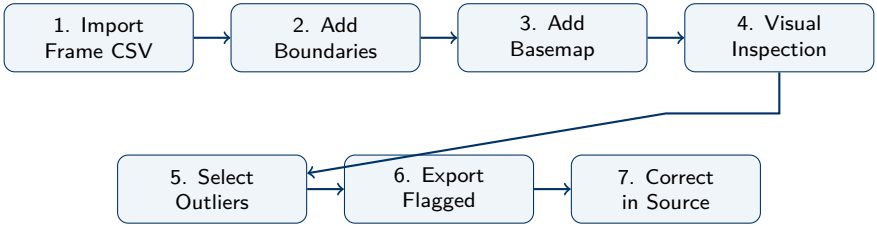
1. Import frame as CSV with coordinates
2. Add administrative boundary layers
3. Add satellite/OpenStreetMap basemap
4. Visually inspect point locations
5. Use selection tools to identify outliers

Key QGIS Features:

- **Geometry Checker** — Automated topology validation
- **Point Sampling Tool** — Extract attributes at locations
- **Spatial Query** — Select points within/outside polygons

Download: https://qgis.org

QGIS: Step-by-Step Validation



Tip: Use Google Satellite or Bing Aerial imagery to verify that points fall on actual settlements, not water or forest.

R Alternative: Interactive Leaflet Maps

```
library(leaflet)

# Create interactive validation map
leaflet(frame_sf) %>%
  addTiles() %>%  # OpenStreetMap basemap
  addCircleMarkers(
    radius = 3,
    color = ~ifelse(coord_valid, "green", "red"),
    popup = ~paste("EA:", ea_id, "<br>",
                   "Province:", province)
  ) %>%
  addPolygons(
    data = country,
    fill = FALSE,
    color = "blue",
    weight = 2
  )

# Opens in browser - zoom, pan, click for details
```

Leaflet maps are shareable as HTML files for team review.

Validation Step 5: Coverage Assessment

Does the frame cover the entire target area?

Check for:

- Districts with no frame units
- Sparse areas (low density)
- New settlements not in frame
- Informal settlements excluded

Methods:

- Point-in-polygon counts
- Density heat maps
- Compare to census counts
- Satellite imagery review

```
# Count frame units per district
frame_counts <- frame_sf %>%
st_join(districts) %>%
group_by(district_name) %>%
summarise(n_units = n())
```

Generating a Frame Quality Report

Standard Metrics to Report:

Metric	Count	%	Status
Total frame units	15,420	100%	—
Valid coordinates	15,112	98.0%	✔
Missing coordinates	185	1.2%	✔
Outside boundaries	43	0.3%	✔
Duplicate locations	80	0.5%	⚠
Districts covered	52/52	100%	✔

Recommendation Thresholds:

- >95% valid coordinates: Proceed with sampling
- 90–95% valid: Clean before sampling
- <90% valid: Major remediation required

Automated Frame QA Script

```
validate_frame <- function(frame, boundary) {
  # Convert to sf
  frame_sf <- st_as_sf(frame,
    coords = c("longitude", "latitude"), crs = 4326)

  # Run checks
  results <- list(
    n_total = nrow(frame),
    n_missing = sum(is.na(frame$latitude)),
    n_outside = sum(!st_within(frame_sf, boundary,
                               sparse = FALSE)),
    n_duplicate = nrow(frame) -
                  n_distinct(paste(frame$latitude,
                                   frame$longitude))
  )

  results$pct_valid <- 100 * (results$n_total -
    results$n_missing - results$n_outside) /
    results$n_total

  return(results)
}
```

Remediation: Fixing Frame Errors

Options by Error Type:

Error Type	Remediation Approach
Missing coordinates	Geocode from address/EA name; manual lookup; exclude
Outside boundaries	Verify against source; correct if typo; exclude if invalid
Swapped lat/long	Automatic swap if pattern clear; manual review
Duplicates	Verify if legitimate; assign unique within-building ID
Wrong CRS	Reproject to correct system

⚠ Document all corrections! Maintain audit trail of frame modifications.

Integrating Validation with Sample Selection

Zambara Case

Lindiwe now has a validated frame. Time to select the sample using spatial methods.

Recommended Workflow:

1. **Validate Frame** — Run QA checks, fix errors
2. **Stratify Spatially** — Province × Urban/Rural
3. **Select Using GRTS** — Ensure spatial balance within strata
4. **Export with Coordinates** — Include lat/long for field teams
5. **Create Maps** — Visual confirmation of coverage

Validated coordinates enable GPS navigation to sample units, reducing listing errors.

Using Coordinates for Field Operations

GPS-Enabled Data Collection:

Benefits:

- Navigate directly to EA
- Verify correct location
- Reduce listing errors
- Track interviewer movements
- Real-time coverage monitoring

Tools:

- Survey Solutions (World Bank)
- ODK/KoboToolbox
- CSPro with GPS
- Custom tablet applications

Export Format for Field:

```
EA_ID, Province, District, Latitude, Longitude
ZA001, Gauteng, Johannesburg, -26.2041, 28.0473
ZA002, Gauteng, Johannesburg, -26.1952, 28.0341
```

Software Options for Spatial Sampling

Task	R	STATA	QGIS	ArcGIS
Load shapefiles	✓ sf	✓ shp2dta	✓	✓
GRTS sampling	✓ spsurvey	✗	✗	✓
Visual validation	✓ leaflet	✗	✓	✓
Coordinate checks	✓	⚠	✓	✓
Choropleth maps	✓ tmap	✓ spmap	✓	✓
Free/Open Source	✓	✗	✓	✗

Recommendation: R for spatial sampling and analysis; QGIS for visual validation and mapping. Both are free.

Application: Validating Zambara's Frame

Zambara Case

Lindiwe applies the validation workflow to ZNSO's new georeferenced EA frame...

Results:

Check	Result	Target	Status
Total EAs	84,907	—	—
Valid coordinates	83,421 (98.2%)	>95%	✓
Missing coordinates	892 (1.1%)	<2%	✓
Outside boundaries	156 (0.2%)	<1%	✓
Duplicate locations	438 (0.5%)	<1%	✓
Districts covered	52/52	100%	✓

Verdict: Frame quality acceptable for sampling.

Module 2.3: Key Takeaways

1. **Spatial Methods** improve coverage when frame has coordinates
2. **Grid-Based Sampling** guarantees minimum spacing
3. **GRTS** provides spatially balanced probability samples
4. **Frame Validation** is essential before sampling
5. **Check for:** Missing coords, outside boundaries, duplicates, swaps
6. **Use GIS Tools:** R (sf, spsurvey, leaflet) and QGIS for validation
7. **Document Quality** in methodology report

A validated, georeferenced frame enables both better sampling and better fieldwork.

End of Module 2.3: GIS Integration

Module 2.3 Complete

GIS Fundamentals • Coverage Analysis • Spatial Sampling

Frame Validation • QGIS/R Tools

Up Next: Module 2.4 — AI/ML Applications

- Machine learning for non-response prediction
- Satellite imagery for frame updates
- Ethical considerations

Break: 15 minutes — Lab 2.3 exercises available

Module 2.4

AI/ML Applications in Sampling

Beyond the Hype: What Can Machine Learning Actually Do?

Slides 211–240

Lindiwe's Question

Zambara Case

After mapping coverage gaps, Lindiwe turns to her colleague Dr. Moyo, the data scientist. "Everyone talks about AI. Can machine learning help us predict which households will refuse to participate?"

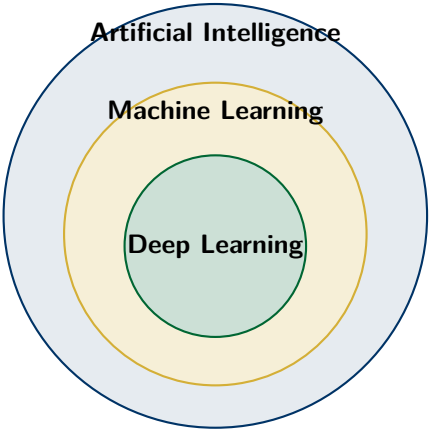
"Can we predict who will say no?"

If we could predict non-response:

- Assign best interviewers to difficult cases
- Plan more contact attempts for high-risk households

The Buzzword Problem

Terms often used interchangeably (but shouldn't be):



Broad field: machines that "think" or act intelligently

Algorithms that learn patterns from data

Neural networks with many layers

For survey sampling, we mostly use **Machine Learning** — not deep learning or "general AI."

What is Machine Learning?

A Simple Definition:

> **Machine Learning** = Algorithms that improve their performance through experience (data)

Key Idea: Instead of programming explicit rules, we let the algorithm *discover* patterns in data.

Example:

- **Traditional:** "If income > R20,000 AND urban, then high refusal risk"
- **ML:** "Here's data on 10,000 households. Learn what predicts refusal."

Machine Learning vs. Traditional Statistics

Aspect	Traditional Statistics	Machine Learning
Primary Goal	Inference & explanation	Prediction
Model Selection	Theory-driven	Data-driven
Interpretability	High (coefficients)	Often low ("black box")
Sample Size	Works with small n	Needs large n
Assumptions	Explicit (normality, etc.)	Fewer assumptions
Overfitting Risk	Lower	Higher (needs regularization)

They're complementary, not competitors. Use statistics for understanding *why*; use ML for predicting *what*.

Plot Twist: Regression IS Machine Learning

⚠ Linear and logistic regression are machine learning algorithms!

The ML family includes:

- Linear regression (oldest ML algorithm!)
- Logistic regression
- Decision trees
- Random forests
- Support vector machines
- Neural networks
- ...and many more

"ML" often just means using algorithms beyond basic regression, with focus on prediction over inference.

Types of Machine Learning

Supervised Learning

Learn from labeled examples
(input → known output)

Example: Predict refusal
from household characteristics

Unsupervised Learning

Find patterns without labels
(clustering, dimensionality)

Example: Group similar
households for stratification

Reinforcement Learning

Learn by trial and error
(agent ↔ environment)

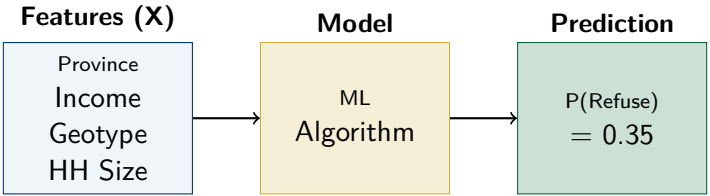
Example: Optimize
contact sequences

For non-response prediction, we use **Supervised Learning** — we have labeled data (responded/refused).

Non-Response Prediction as a Classification Problem

Setup:

- **Input (X):** Household characteristics from frame
- **Output (Y):** Binary — Responded (0) or Refused (1)
- **Goal:** Learn function $f(X) \to Y$



The model outputs a **probability** of refusal, which we can use to prioritize follow-up efforts.

Logistic Regression: The Classic Approach

Model:

$$\log\left(\frac{P(\text{Refuse})}{1-P(\text{Refuse})}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots$$

Strengths:

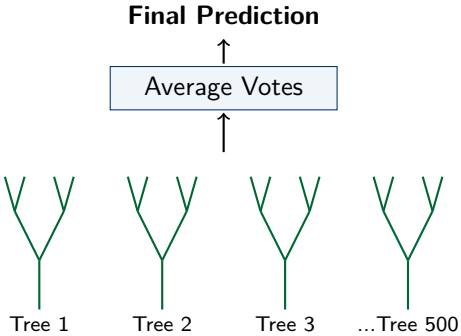
- Coefficients are interpretable (odds ratios)
- Statistical inference (p-values, confidence intervals)
- Works well with smaller samples

Limitations:

- Assumes linear relationship (in log-odds)
- Cannot capture complex interactions automatically
- May underfit if relationships are non-linear

Random Forest: A Modern Alternative

Concept: Build many decision trees, each on a random subset of data, then average their predictions.



Key Advantage: Captures non-linear relationships and interactions automatically.

Lab 2.4: Building a Non-Response Prediction Model

Zambara Case

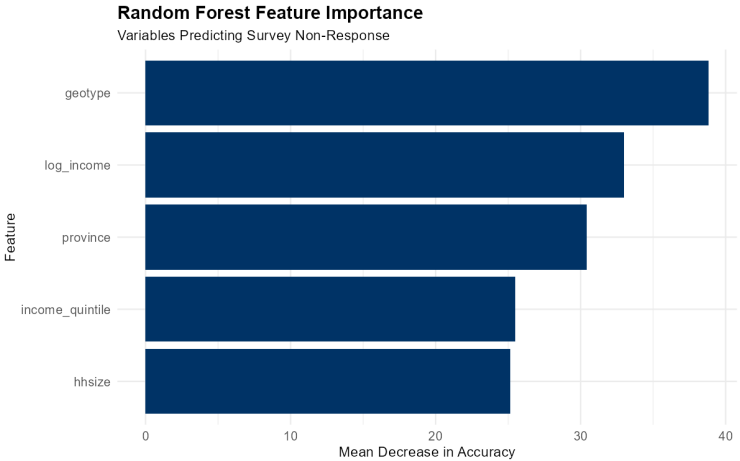
Dr. Moyo helps Lindiwe build a Random Forest model using GHS 2024 data to predict which households might refuse participation.

Setup:

- **Data:** GHS 2024 household file (20,940 observations)
- **Target:** Simulated non-response (based on realistic patterns)
- **Features:** Province, geotype, income, household size
- **Model:** Random Forest with 500 trees
- **Split:** 70% training, 30% testing

Software: R with `randomForest` and `caret` packages

Lab 2.4 Results: Feature Importance



Source: Lab 2.4 Random Forest model on GHS 2024 simulated non-response

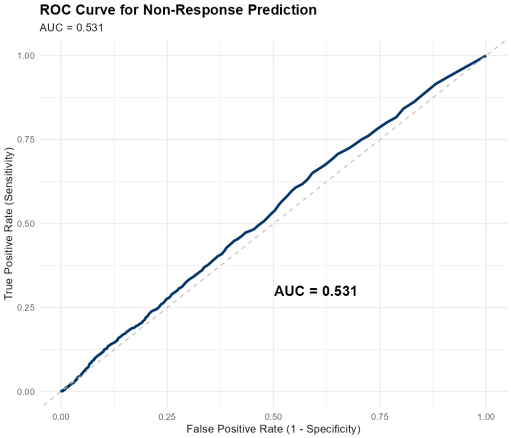
Interpreting Feature Importance

What the model learned — Top predictors of non-response:

1. **Geotype** (Mean Decrease Accuracy: 39)
   - Urban households more likely to refuse
   - Privacy concerns, "too busy," security issues
2. **Log Income** (MDA: 33)
   - Higher income $\rightarrow$ higher refusal probability
   - Opportunity cost of time, privacy sensitivity
3. **Province** (MDA: 30)
   - Regional variation in survey culture
   - Gauteng has historically higher refusal rates

These findings align with survey methodology literature on non-response correlates.

Lab 2.4 Results: Model Performance (ROC Curve)



Source: Lab 2.4 — ROC curve on 30% test set

The Reality Check: AUC = 0.531

⚠ AUC = 0.531

Barely better than random guessing (AUC = 0.50)

Zambara Case

Lindiwe looks disappointed. "So the model doesn't really work?" Dr. Moyo smiles. "This is actually a valuable lesson about AI in the real world."

AUC Interpretation Scale:

- 0.50 = Random guessing (useless)
- 0.60–0.70 = Poor discrimination
- 0.70–0.80 = Acceptable

Why Non-Response Prediction is Hard

Non-response depends on factors we *cannot* observe in the frame:

What We Know (Frame):

- Province
- Urban/rural
- EA characteristics
- (Maybe) dwelling type

What Actually Predicts Refusal:

- Personality/attitudes
- Previous survey experience
- Trust in government
- Current life circumstances
- Who answers the door

The predictable component of non-response is small. Most variation is "random" from the model's perspective.

AI in Surveys: Hype vs. Reality

The Hype	The Reality
"AI will predict exactly who refuses"	Models achieve modest accuracy at best
"ML replaces survey expertise"	ML is a tool, not a replacement
"More data = better predictions"	Garbage in, garbage out
"Black box models are fine"	Interpretability matters for trust
"AI eliminates non-response bias"	AI can help, but cannot eliminate

Zambara Case

Dr. Moyo explains: "AI is powerful, but it's not magic. For survey work, it's most useful for *incremental* improvements, not revolutionary change."

Where ML Actually Helps in Surveys

Realistic Applications:

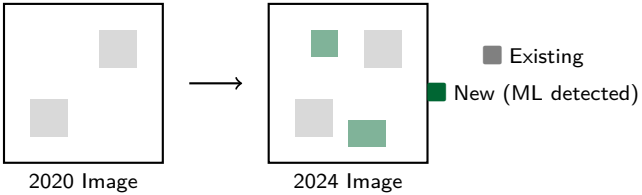
1. **Prioritization** (even weak predictions help)
   - Focus resources on highest-risk cases first
   - Small improvements in efficiency add up
2. **Frame Enhancement**
   - Satellite imagery to detect new dwellings
   - Administrative data linkage
3. **Data Quality Checks**
   - Anomaly detection in responses
   - Interviewer performance monitoring
4. **Small Area Estimation**
   - Combine survey data with auxiliary variables
   - Predictions for unsampled areas

ML Application: Satellite Imagery for Frame Updates

Problem: Sampling frames become outdated between censuses.

ML Solution: Use satellite/aerial imagery + deep learning to detect:

- New construction (buildings that didn't exist in frame)
- Demolished structures
- Informal settlements
- Changes in building density



Source: WorldPop, Facebook AI Research, and various NSO initiatives

Ethical Considerations in AI for Surveys

With great power comes great responsibility

Key Questions:

- Who benefits from the predictions?
- Who might be harmed?
- Is the model fair across groups?
- What happens when the model is wrong?

Ethical Issue 1: Algorithmic Bias

Problem: ML models can perpetuate or amplify existing biases.

Example Scenario:

- Model predicts high refusal for low-income households
- Field managers assign fewer resources to these areas
- Self-fulfilling prophecy: under-resourced areas have higher non-response
- Low-income populations become systematically under-represented

Mitigation:

- Audit models for disparate impact across groups
- Use predictions to *increase* effort, not decrease it
- Monitor outcomes by demographic characteristics

Ethical Issue 2: Privacy and Consent

Problem: More data improves predictions, but raises privacy concerns.

Questions to Consider:

- Did respondents consent to their data being used for ML?
- What if we link survey data to administrative records?
- Can individual households be identified from predictions?
- Who has access to the trained model?

Best Practices:

- Clear data governance policies
- Aggregate predictions (not individual-level sharing)
- Regular privacy impact assessments
- Compliance with data protection regulations (POPIA, GDPR)

Ethical Issue 3: Transparency and Explainability

Problem: "Black box" models make decisions we can't explain.

Stakeholder Questions:

- "Why was my area flagged high-risk?"
- "How do you know the model is fair?"
- "What if the model makes mistakes?"

Solutions:

- Use interpretable models when possible
- Document model development
- Provide feature importance
- Enable human override

For NSOs, **trust** is paramount. Unexplainable AI can undermine public confidence in official statistics.

Best Practices for AI in National Statistics Offices

Recommendations:

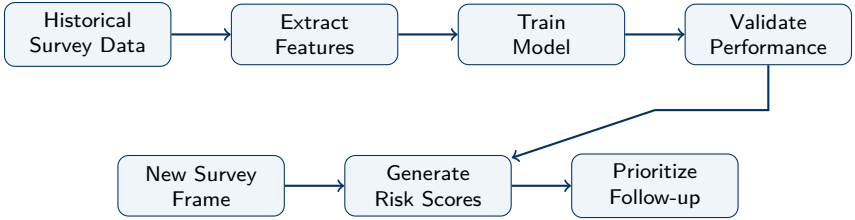
1. **Start Simple**
   - Begin with interpretable models (logistic regression)
   - Add complexity only if justified by performance gains
2. **Validate Rigorously**
   - Always hold out test data
   - Check performance across subgroups
3. **Document Everything**
   - Model development process
   - Training data sources
   - Known limitations
4. **Human in the Loop**
   - ML informs decisions, doesn't make them
   - Enable override by domain experts

Lab 2.4: Comparing Approaches

Aspect	STATA (Logistic)	R (Random Forest)
Model Type	Parametric	Non-parametric
Output	Odds ratios, p-values	Variable importance
Interpretation	Direct coefficient meaning	Relative importance only
Assumptions	Linearity in log-odds	None
Prediction	Moderate	Often better
Computation	Fast	Slower (500 trees)
Best For	Understanding "why"	Maximizing prediction

Recommendation: Use logistic regression for explanation, Random Forest for prediction. Compare both!

Practical Workflow for Non-Response Prediction



Key Point: Model is trained on *past* survey data, then applied to *new* survey frame before fieldwork begins.

Lindiwe's Conclusion

Zambara Case

Lindiwe summarizes her learnings for the ZNSO team meeting...

"Here's what I learned about AI for surveys:"

1. ML is **not magic** — it's sophisticated statistics
2. Non-response prediction is **hard** — even good models have limits
3. The value is in **prioritization**, not perfect prediction
4. **Interpretability** matters for trust and accountability
5. **Ethics** must guide how we use predictions
6. Start **simple**, add complexity only when justified

Module 2.4: Key Takeaways

1. **ML $\approx$ advanced statistics** — regression is ML too
2. **Supervised learning** for prediction problems with labeled data
3. **Random Forest** captures non-linear patterns automatically
4. **AUC** measures discrimination ability (0.5 = random, 1.0 = perfect)
5. **Non-response is hard to predict** — most variation is "random"
6. **Useful applications:** frame updates, prioritization, quality checks
7. **Ethics matter:** bias, privacy, transparency

AI is a tool to enhance survey methodology, not replace human expertise.

Discussion Questions

For your NSO context:

1. What data do you have from *past* surveys that could train a non-response model?
2. What **ethical concerns** might arise from using ML predictions in your country?
3. How would you **explain** an ML-based decision to a skeptical stakeholder?
4. What **other applications** of ML might benefit your survey operations?
5. How can you balance **innovation** with maintaining public trust?

Group Discussion: 10 minutes

End of Module 2.4 & Day 2

Day 2 Complete

Computational Tools • R/STATA • GIS Integration
Spatial Sampling • AI/ML Applications

Tomorrow — Day 3: Advanced Weighting and Variance Estimation

- Base weight development
- Calibration and raking
- Taylor linearization and replication methods

Thank you! See you tomorrow at 9:00 AM

Predictive Stratification

From Prediction to Action

Optimizing Field Operations with Risk Scores

Module 2.4 Continued — Slides 241–270

From Prediction to Action

Zambara Case

Lindiwe has her risk scores from Lab 2.4. Now the Field Operations Manager, Mr. Dlamini, asks: "How do I actually *use* these predictions?"

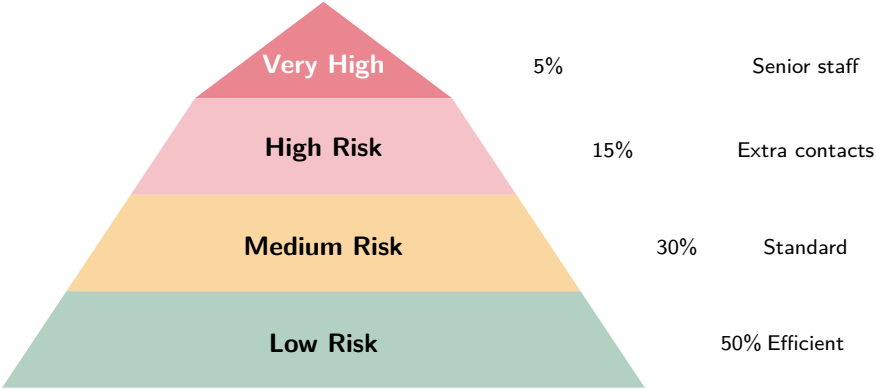
"What do we DO with AUC = 0.531?"

Key Insight: Even weak predictions can improve operations if used wisely.

- We don't need perfect prediction
- We need *better than random* resource allocation
- Small efficiency gains at scale = big impact

Risk Stratification: The Core Idea

Divide households into risk tiers based on predicted refusal probability:



Different risk tiers receive different treatment protocols.

Lab 2.4 Results: Risk Category Distribution

From our Random Forest model:

Risk Category	P(Refuse)	Households	%	Actual Refusal
Low	<15%	8,450	40.4%	11.2%
Medium	15–25%	7,120	34.0%	18.5%
High	25–35%	3,890	18.6%	27.3%
Very High	>35%	1,480	7.1%	38.1%
Total	—	**20,940**	100%	17.8%

Model correctly identifies that Very High risk tier has 3.4× higher refusal rate than Low risk tier.

The Value of Predictive Stratification

Zambara Case

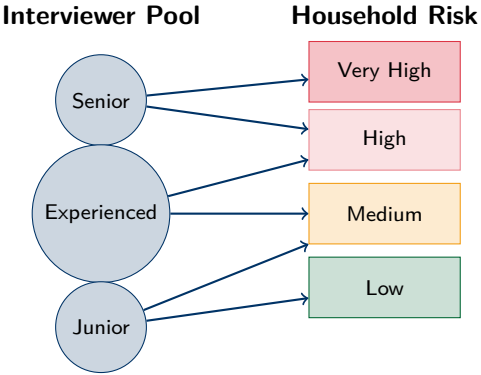
Mr. Dlamini realizes: "Even if we can't predict *exactly* who refuses, knowing that 38% of the 'Very High' group will refuse vs. only 11% of 'Low' is actionable!"

What Stratification Enables:

1. **Targeted Resource Allocation**
   - More effort where it's needed most
2. **Differentiated Protocols**
   - Custom approaches for different risk levels
3. **Realistic Expectations**
   - Plan for higher non-response in certain areas
4. **Efficiency Gains**
   - Don't over-invest in low-risk cases

Smart Interviewer Assignment

Match interviewer experience to household risk:



Rationale: Senior interviewers have better refusal conversion skills.

Implementing Smart Assignment in R

```r
# Assign interviewers based on risk category
assign_interviewer <- function(risk_category, interviewer_pool) {

  # Priority rules
  if (risk_category == "Very High") {
    return(sample(interviewer_pool$senior, 1))

  } else if (risk_category == "High") {
    pool <- c(interviewer_pool$senior, interviewer_pool$experienced)
    return(sample(pool, 1))

  } else if (risk_category == "Medium") {
    return(sample(interviewer_pool$experienced, 1))

  } else {  # Low risk
    return(sample(interviewer_pool$junior, 1))
  }
}

# Apply to all households
sample_data$assigned_interviewer <- sapply(
  sample_data$risk_category,
```

Differentiated Contact Strategies

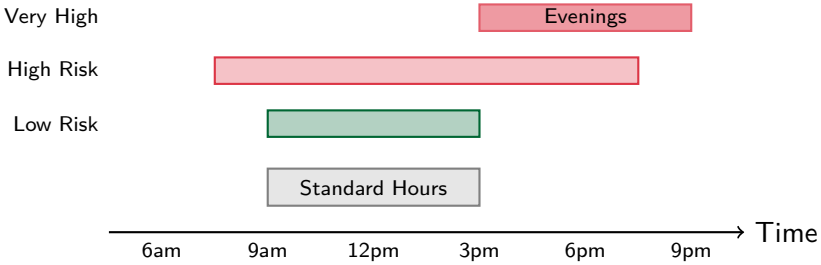
Customize contact attempts by risk level:

Parameter	Low	Medium	High	Very High
Max contact attempts	3	4	6	8
Days between attempts	3	2	1	1
Evening/weekend visits	No	Optional	Yes	Required
Advance letter	No	No	Yes	Yes
Incentive offered	No	No	Optional	Yes
Supervisor callback	No	No	Optional	Required

High-risk households get more intensive protocols from the start, not as a last resort.

Optimizing Contact Timing

High-risk households often require non-standard timing:

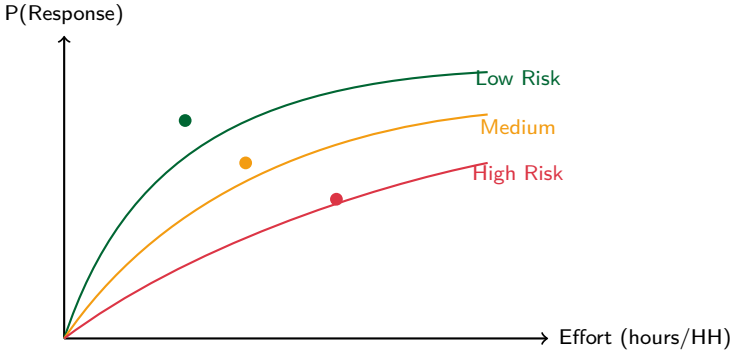


High-income urban households: Often only available evenings/weekends.

Resource Allocation Optimization

Budget constraint: Fixed interviewer-hours per region.

Question: How to allocate effort across risk tiers?



Diminishing returns: Optimal allocation is NOT equal effort per household.

Workload Balancing Across Teams

Problem: Some areas have more high-risk households.

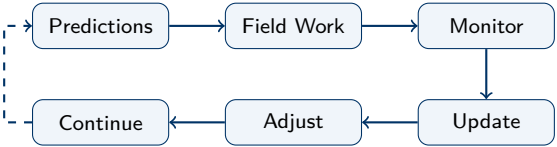
Team	HH Assigned	Avg Risk	Est. Hours	Adjusted
Team A (Urban)	150	High	450	180 HH
Team B (Rural)	150	Low	250	120 HH
Team C (Mixed)	150	Medium	350	150 HH

Solution: Adjust workload assignments based on predicted difficulty.

- High-risk areas: Fewer households per interviewer
- Low-risk areas: More households per interviewer
- Equal *expected effort*, not equal *count*

Adaptive Survey Design

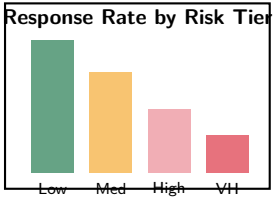
Definition: Modifying survey protocols *during* data collection based on emerging information.

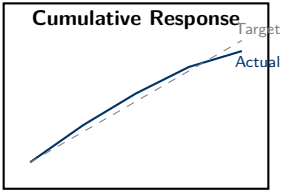


Key: Learn from early results, improve predictions, adapt strategy.

Real-Time Monitoring Dashboard

Track predicted vs. actual performance:



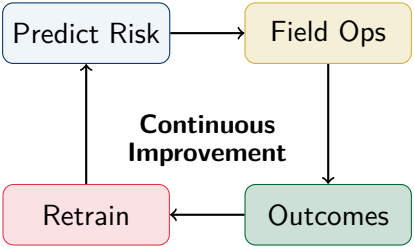


Dashboard Metrics:

- Response rates by predicted risk category
- Model calibration (predicted vs. actual)
- Resource utilization by team
- Cases requiring escalation

Model Improvement Feedback Loop

Use field outcomes to improve future predictions:



Each survey wave improves predictions for the next!

Implementation: Zambara Case Study

Zambara Case

ZNSO implements predictive stratification for GHS 2025 field operations...

Phase 1: Pre-Fieldwork

- Train model on GHS 2024 non-response patterns
- Score all 21,000 sample households
- Stratify into 4 risk categories

Phase 2: Assignment

- Match 45 senior interviewers to 1,480 Very High risk HH
- Assign differentiated protocols by tier
- Adjust workloads for regional risk variation

Phase 3: Monitoring

- Daily dashboard tracking predicted vs. actual

Expected Benefits

Projected improvements from predictive stratification:

Metric	Traditional	With ML
Overall response rate	82%	85% (+3%)
Very High risk response	62%	70% (+8%)
Contact attempts (avg)	4.2	3.8 (-10%)
Interviewer hours/complete	2.1	1.9 (-10%)
Non-response bias (est.)	Moderate	Reduced

Even modest improvements (3% response rate) can significantly reduce non-response bias.

Cost-Benefit Analysis

Is predictive stratification worth the investment?

Costs:

- Data scientist time (model building)
- Computing infrastructure
- Training for field managers
- Dashboard development
- Ongoing maintenance

Estimate: R150,000/year

Benefits:

- Reduced fieldwork costs
- Higher response rates
- Lower non-response bias
- Better quality estimates
- Reusable for future surveys

Estimate: R400,000/year savings

ROI: Approximately 2.7:1 for large continuous surveys.

Implementation Challenges

Real-world obstacles:

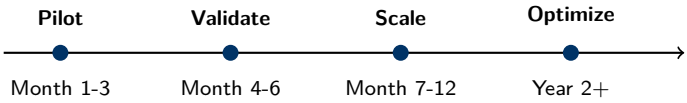
1. **Data Availability**
   - Historical non-response data may be incomplete
   - Frame variables limited for prediction
2. **Organizational Resistance**
   - "We've always done it this way"
   - Skepticism about "black box" models
3. **Technical Capacity**
   - Need staff with ML skills
   - IT infrastructure requirements
4. **Operational Complexity**
   - Integrating predictions into existing workflows
   - Training field supervisors

Strategies to Overcome Challenges

Challenge	Strategy
Limited historical data	Start with simple models; improve over time
Organizational resistance	Pilot project with measurable outcomes
Technical capacity	Partner with university/research institute
Workflow integration	Design dashboard for field managers, not statisticians
Model trust	Emphasize interpretable models; show feature importance

Start small, demonstrate value, then scale. Don't try to revolutionize everything at once.

Phased Implementation Roadmap



Phase Details:

- **Pilot (1-3):** Build model, test on subset
- **Validate (4-6):** Evaluate results, refine approach
- **Scale (7-12):** Full deployment, train staff
- **Optimize (2+):** Continuous improvement

Predictive Stratification in the SADC Context

Opportunities for SADC NSOs:

- **Shared Learning**
  - Cross-country model development
  - Regional workshops on ML applications
- **Common Challenges**
  - High non-response in urban areas
  - Informal settlements
  - Mobile populations
- **Donor Support**
  - World Bank, UNFPA interested in innovation
  - Potential for regional pilot funding
- **Technical Partnerships**
  - Regional statistics centers
  - University collaborations

Emerging Data Sources: Mobile Phone Metadata

Future possibility: Predict non-response using mobile phone patterns.

Potential Predictors:

- Call/SMS frequency
- Mobility patterns
- Top-up behavior
- Device type
- Home location estimate

Challenges:

- Privacy concerns (major)
- Data access agreements
- Not everyone has mobile
- Regulatory approval

⚠ Highly experimental. Privacy and ethical review essential before any implementation.

Application: Incentive Optimization

Use predictions to optimize incentive spending:

Risk Category	Base Incentive	Conditional	Expected Cost
Low	R0	R0	R0
Medium	R0	R50 if needed	R15
High	R50	R100 if needed	R65
Very High	R100	R200 if needed	R150

Strategy: Offer larger incentives *upfront* to high-risk households, rather than escalating after refusal.

Proactive incentives are more effective than reactive ones.

Integrating with Quality Assurance

Predictions help target QA resources:

- **Re-interview Selection**
  - Oversample re-interviews from high-risk completed cases
  - Check for falsification in "easy" completions
- **Supervisor Accompaniment**
  - Prioritize observation visits for high-risk assignments
- **Audio Recording**
  - Focus review on high-risk interviews
- **Paradata Analysis**
  - Flag suspiciously short interviews in high-risk areas

Risk-based QA is more efficient than random QA.

Documentation for Transparency

What to document when using predictive stratification:

1. **Model Specification**
   - Algorithm used, hyperparameters
   - Training data source and period
2. **Performance Metrics**
   - AUC, accuracy, calibration plots
   - Performance by subgroup
3. **Operational Use**
   - How predictions affected assignments
   - Protocol differences by risk tier
4. **Outcomes**
   - Predicted vs. actual response rates
   - Impact on non-response bias

Include in survey methodology report for user transparency.

Lindiwe's Field Operations Manual Update

Zambara Case

Lindiwe drafts new guidelines for the GHS 2025 field manual...

Section 7.3: Risk-Based Field Protocols

7.3.1 Each sample household is assigned a risk category (Low/Medium/High/Very High) based on predicted refusal probability.

7.3.2 Interviewers shall follow the differentiated contact protocol specified in Table 7.1.

7.3.3 Very High risk cases shall be assigned to interviewers with at least 2 years experience and a track record of successful refusal conversion.

7.3.4 Supervisors shall review all Very High risk non-contacts within 48 hours and may authorize additional measures.

Future Developments: Where Is This Heading?

Near-term (1–3 years):

- Integration with CAPI systems (automatic protocol assignment)
- Real-time model updating during fieldwork
- Cross-survey learning (LFS, census, GHS share models)

Medium-term (3–5 years):

- Respondent-level (not just household) prediction
- Natural language processing for interviewer notes
- Automated interviewer scheduling optimization

Long-term (5+ years):

- Fully adaptive surveys (AI adjusts in real-time)
- Integrated frame-survey-admin data ecosystems
- Predictive sampling (not just non-response)

Predictive Stratification: Key Takeaways

1. **Even weak predictions help** — AUC = 0.53 is still useful for prioritization
2. **Stratify by risk** — Different tiers get different protocols
3. **Match resources to difficulty** — Senior staff for hard cases
4. **Optimize contact strategies** — More attempts, better timing for high-risk
5. **Balance workloads** — Adjust for predicted difficulty, not just count
6. **Monitor and adapt** — Use outcomes to improve predictions
7. **Document everything** — Transparency builds trust

Predictive stratification is **operational optimization**, not magic.

Exercise: Design Your Stratification Protocol

Group Activity (15 minutes):

Using the Lab 2.4 risk scores for your country context:

1. Define **4 risk tiers** and their probability thresholds
2. Specify **contact protocols** for each tier:
   - Maximum contact attempts
   - Timing requirements
   - Incentive policy
3. Design **interviewer assignment** rules
4. Identify **3 metrics** for your monitoring dashboard
5. List **2 potential obstacles** in your NSO context

Present: Each group shares their protocol design (3 min each)

End of Day 2: Computational Tools Complete

Day 2 Complete

Module 2.1: Software Overview
Module 2.2: PPS Sampling
Module 2.3: GIS Integration
Module 2.4: AI/ML Applications

Day 3 Preview: Advanced Weighting & Variance Estimation

Thank you! See you tomorrow at 9:00 AM

Evening: Optional lab session for hands-on practice

Day 2 Synthesis

Cheat Sheet 2.0 & Homework

Computational Tools for Survey Sampling

Slides 271–300

Day 2: The Journey

Zambara Case

Lindiwe started the day wondering how to use modern tools for sampling. By evening, she has built spatial maps, predicted non-response, and designed smart field protocols.

What We Covered:

Morning

- Software ecosystem overview
- R `survey` package deep dive
- STATA `svy` commands
- PPS systematic sampling

Afternoon

- GIS integration fundamentals
- Spatial sampling methods
- AI/ML for non-response
- Predictive stratification

Learning Objectives: Self-Assessment

Can you now...

- [x] Configure survey design objects in R and STATA?
- [x] Implement PPS systematic sampling using software?
- [x] Load and join spatial data with survey datasets?
- [x] Create choropleth maps showing survey estimates?
- [x] Identify spatial coverage gaps in your sample?
- [x] Explain what machine learning can (and cannot) do?
- [x] Design risk-stratified field protocols?

If any boxes are unchecked, review the relevant lab scripts tonight!

Cheat Sheet 2.0

Essential Commands for Survey Sampling

R & STATA Side-by-Side

Cheat Sheet: Survey Design Setup

R (survey package)

```r
library(survey)

# Basic design
des <- svydesign(
  id = ~psu,
  strata = ~stratum,
  weights = ~house_wgt,
  data = mydata,
  nest = TRUE
)

# Check design
summary(des)
```

STATA

```stata
* Basic design
svyset psu [pw=house_wgt], ///
  strata(stratum)

* Check design
svydescribe

* View settings
svyset
```

⚠ Always set up design BEFORE computing estimates!

Cheat Sheet: Point Estimates

R

```r
# Mean
svymean(~income, des, na.rm=TRUE)

# Total
svytotal(~income, des, na.rm=TRUE)

# Proportion (factor)
svymean(~factor(employed), des)

# Quantiles
svyquantile(~income, des,
            quantiles=c(0.25,0.5,0.75))

# Ratio
svyratio(~income, ~hhsize, des)
```

STATA

```stata
* Mean
svy: mean income

* Total
svy: total income

* Proportion
svy: proportion employed

* Quantiles (requires epctile)
epctile income [pw=house_wgt], ///
  p(25 50 75)

* Ratio
svy: ratio income/hhsize
```

Cheat Sheet: Subgroup (Domain) Analysis

R

```r
# By province
svyby(~income, ~province, des,
      svymean, na.rm=TRUE)

# By multiple groups
svyby(~income, ~province + geotype,
      des, svymean)

# Subset (correct way!)
urban <- subset(des, geotype==1)
svymean(~income, urban)
```

STATA

```stata
* By province
svy: mean income, over(province)

* By multiple groups
svy: mean income, ///
  over(province geotype)

* Subset (subpop is correct!)
svy, subpop(if geotype==1): ///
  mean income
```

⚠ Never filter data before analysis — use `subset()` in R or `subpop()` in STATA!

Cheat Sheet: Variance Estimation

R

```r
# Standard errors (default)
svymean(~income, des)

# Confidence intervals
confint(svymean(~income, des))

# Design effect
svymean(~income, des, deff=TRUE)

# Coefficient of variation
cv(svymean(~income, des))
```

STATA

```stata
* Standard errors (default)
svy: mean income

* Confidence intervals
svy: mean income
estat ci

* Design effect
estat effects

* Coefficient of variation
estat cv
```

Default: Taylor linearization. Day 3 covers replication methods (JK, BRR).

Cheat Sheet: Regression with Survey Weights

R

```r
# Linear regression
model <- svyglm(income ~ education +
                age + female,
                design = des)
summary(model)

# Logistic regression
logit <- svyglm(employed ~ education,
                design = des,
                family = quasibinomial())
```

STATA

```stata
* Linear regression
svy: regress income education ///
  age female

* Logistic regression
svy: logit employed education

* Margins
margins, dydx(*)
```

In R, use `quasibinomial()` for binary outcomes to avoid overdispersion warnings.

Cheat Sheet: Loading Spatial Data

R (sf package)

```r
library(sf)

# Load shapefile
shp <- st_read("boundaries.shp")

# Check CRS
st_crs(shp)

# Transform CRS
shp <- st_transform(shp, 4326)

# View attributes
head(shp)
plot(shp["province"])
```

STATA

```stata
* Convert shapefile to STATA
shp2dta using boundaries, ///
  database(geo_db) ///
  coordinates(geo_coord) ///
  replace

* Load database
use geo_db, clear
describe

* For mapping, need spmap
ssc install spmap
```

Cheat Sheet: Joining Survey to Spatial Data

R

```r
library(dplyr)

# Aggregate survey data
prov_est <- svyby(~income, ~province,
                  des, svymean) %>%
  rename(prov_code = province)

# Join to shapefile
shp_data <- shp %>%
  left_join(prov_est,
            by = "prov_code")

# Check join
sum(is.na(shp_data$income))
```

STATA

```stata
* Aggregate survey data
collapse (mean) income ///
  [pw=house_wgt], by(province)
rename province prov_code
save prov_est, replace

* Join to geo database
use geo_db, clear
merge 1:1 prov_code using prov_est

* Check join
tab _merge
```

⚠ Ensure join keys match exactly (spelling, case, codes).

Cheat Sheet: Creating Choropleth Maps

R (tmap)

```r
library(tmap)

tm_shape(shp_data) +
  tm_polygons("income",
    style = "quantile",
    palette = "Blues",
    title = "Mean Income") +
  tm_layout(
    main.title = "Income by Province",
    legend.outside = TRUE)

# Save map
tmap_save(filename = "map.png")
```

STATA (spmap)

```stata
* Basic choropleth
spmap income using geo_coord, ///
  id(prov_code) ///
  fcolor(Blues) ///
  title("Income by Province")

* Save map
graph export "map.png", replace
```

R alternative: `ggplot2 + geom_sf()` for more customization.

Cheat Sheet: Random Forest for Prediction (R)

```
library(randomForest)
library(caret)

# Prepare data
model_data <- data %>%
  select(target, feature1, feature2, feature3) %>%
  mutate(target = factor(target)) %>%
  na.omit()

# Train-test split
set.seed(42)
idx <- createDataPartition(model_data$target, p=0.7, list=FALSE)
train <- model_data[idx, ]
test <- model_data[-idx, ]

# Train model
rf <- randomForest(target ~ ., data=train, ntree=500, importance=TRUE)

# Feature importance
importance(rf)
varImpPlot(rf)
```

Cheat Sheet: Logistic Regression for Prediction (STATA)

```
* Prepare data
generate target = (outcome > threshold)
label define targetlbl 0 "No" 1 "Yes"
label values target targetlbl

* Train-test split (70/30)
set seed 42
generate rand = runiform()
generate train = (rand <= 0.70)

* Fit logistic regression on training data
logit target feature1 feature2 feature3 if train==1

* Display odds ratios
logit, or

* Predict probabilities
predict prob_pred, pr

* Evaluate on test set
estat classification if train==0
```

Quick Reference: R vs STATA Commands

Task	R	STATA
Set design	`svydesign()`	`svyset`
Mean	`svymean()`	`svy: mean`
Total	`svytotal()`	`svy: total`
Proportion	`svymean(~factor())`	`svy: proportion`
By group	`svyby()`	`over()`
Subset	`subset(design)`	`subpop()`
Regression	`svyglm()`	`svy: regress`
Design effect	`deff=TRUE`	`estat effects`
Load shapefile	`st_read()`	`shp2dta`
Choropleth	`tm_polygons()`	`spmap`
Random Forest	`randomForest()`	(use R)

Full cheat sheet available in workshop materials folder.

Common Errors and Solutions

Error	Solution
"Lonely PSU" / Single PSU in stratum	Use `options(survey.lonely.psu="adjust")` in R or `svyset, singleunit(certainty)` in STATA
Wrong SEs after filtering	Use `subset()` or `subpop()`, not data filtering
Missing values in estimates	Add `na.rm=TRUE` (R) or check for missings first
CRS mismatch in spatial join	Transform both layers to same CRS before joining
Join produces all NA	Check key variable names, types, and values match exactly
Memory error with large data	Process in chunks or use `data.table/fread()`

Homework

The Spatial Challenge

Map Your Country's Sample

Homework: The Spatial Challenge

Objective: Create a spatial visualization of your country's most recent household survey sample.

Deliverable: A choropleth map showing survey estimates by administrative region, with sample coverage indicators.

Why This Matters:

- Practice the GIS skills learned today
- Identify potential coverage gaps in your own surveys
- Build reusable code for your NSO
- Share insights with fellow participants tomorrow

Time Required: 1–2 hours

The Spatial Challenge: Data You Need

1. **Survey Data (from your NSO):**
   - Recent household survey (HH or person file)
   - Must include: region/province code, weights, key indicator
   - Examples: DHS, MICS, LFS, Census sample

2. **Shapefile (administrative boundaries):**
   - Download from GADM: `https://gadm.org/download_country.html`
   - Select your country, Level 1 (provinces) or Level 2 (districts)
   - Format: Shapefile (.shp)

3. **Join Key:**
   - Identify the variable that links survey to shapefile
   - Usually: province code, region name, or admin code

The Spatial Challenge: Step-by-Step

Step 1: Load and Explore Data

- Load your survey data
- Set up survey design (weights, strata, PSU)
- Compute regional estimates for one indicator

Step 2: Load Shapefile

- Download GADM shapefile for your country
- Load into R (`st_read`) or STATA (`shp2dta`)
- Check the join key variable

Step 3: Join and Map

- Join survey estimates to shapefile
- Create choropleth map
- Add title, legend, and source citation

Step 4: Coverage Analysis

- Count PSUs per region

The Spatial Challenge: R Starter Code

```
# ============================================
# SPATIAL CHALLENGE - YOUR COUNTRY
# Replace placeholders with your data
# ============================================

library(haven); library(survey); library(sf); library(tmap); library(dplyr)

# 1. Load YOUR survey data
mydata <- read_dta("YOUR_SURVEY.dta")

# 2. Set up survey design
des <- svydesign(id = ~YOUR_PSU, strata = ~YOUR_STRATUM,
                 weights = ~YOUR_WEIGHT, data = mydata)

# 3. Compute regional estimates
regional <- svyby(~YOUR_INDICATOR, ~YOUR_REGION, des, svymean, na.rm=TRUE)

# 4. Load YOUR country shapefile
shp <- st_read("gadm41_XXX_1.shp")  # Replace XXX with country code

# 5. Join and map
shp_data <- shp %>% left_join(regional, by = c("NAME_1" = "YOUR_REGION"))

tm_shape(shp_data) +
  tm_polygons("YOUR_INDICATOR", title = "Your Indicator") +
  tm_layout(main.title = "YOUR COUNTRY: Survey Estimates by Region")
```

The Spatial Challenge: STATA Starter Code

```
* ============================================
* SPATIAL CHALLENGE - YOUR COUNTRY
* Replace placeholders with your data
* ============================================

* 1. Load YOUR survey data
use "YOUR_SURVEY.dta", clear

* 2. Set up survey design
svyset YOUR_PSU [pw=YOUR_WEIGHT], strata(YOUR_STRATUM)

* 3. Compute regional estimates
svy: mean YOUR_INDICATOR, over(YOUR_REGION)
matrix results = r(table)

* Or collapse for mapping
preserve
collapse (mean) YOUR_INDICATOR [pw=YOUR_WEIGHT], by(YOUR_REGION)
save regional_est, replace
restore

* 4. Convert shapefile
shp2dta using gadm41_XXX_1, database(geo_db) coordinates(geo_coord) replace

* 5. Merge and map
use geo_db, clear
merge 1:1 NAME_1 using regional_est
spmap YOUR_INDICATOR using geo_coord, id(_ID) fcolor(Blues)
```

The Spatial Challenge: Bonus Points

For extra credit, try these extensions:

1. **Coverage Map** (+10 points)
   - Create a second map showing PSU counts per region
   - Identify regions with fewer than 30 PSUs
2. **Urban/Rural Comparison** (+15 points)
   - Create side-by-side maps for urban vs. rural
   - Use `tmap_arrange()` in R
3. **Coefficient of Variation Map** (+20 points)
   - Map the CV (precision) instead of the estimate
   - Identify regions with high uncertainty
4. **Interactive Map** (+25 points)
   - Use `tmap_mode("view")` for leaflet output
   - Export as HTML for sharing

The Spatial Challenge: Submission

What to Prepare for Tomorrow:

1. **Your Map** (PNG or PDF)
   - Clear title with country and indicator
   - Legend with units
   - Source citation
2. **Your Code** (R script or do-file)
   - Commented and reproducible
   - Shows data loading through final map
3. **Brief Findings** (2–3 sentences)
   - What regional patterns did you find?
   - Any coverage gaps identified?

Sharing: We'll have a 15-minute "Map Gallery" at the start of Day 3!

The Spatial Challenge: Troubleshooting

Common Issues and Solutions:

Join fails:	Check that region codes/names match exactly between survey and shapefile. Try `unique()` on both to compare.
Map is blank:	Verify the joined data has non-NA values. Check `summary(shp_data$indicator)`.
CRS error:	Transform shapefile to WGS84: `st_transform(shp, 4326)`.
Can't find shapefile:	GADM uses ISO country codes. South Africa = ZAF, Botswana = BWA, etc.
Survey design error:	Ensure PSU and stratum variables exist and have no missing values.

Instructors available during evening lab session for help!

Day 3 Preview

Advanced Weighting & Variance Estimation

The Math Returns!

Day 3: What's Coming

The Weight of Evidence

Zambara Case

First field results arrive for Lindiwe's survey. Raw estimates differ dramatically from administrative data. She must develop rigorous weighting and calibration procedures...

Topics:

- Base weight calculation from selection probabilities
- Non-response weight adjustments
- Calibration methods: Raking and GREG
- Variance estimation: Taylor linearization
- Replication methods: Jackknife and BRR

Day 3: Mathematical Foundations

Key Formulas You'll Master:

Base Weight:

$$w_i = \frac{1}{\pi_i} = \frac{1}{P(\text{unit } i \text{ selected})}$$

Calibration Estimator:

$$\hat{Y}_{cal} = \sum_{i \in s} g_i w_i y_i \quad \text{where} \quad \sum_{i \in s} g_i w_i \mathbf{x}_i = \mathbf{X}$$

Taylor Linearization Variance:

$$\widehat{\text{Var}}(\hat{\theta}) \approx \sum_{h=1}^{H} \frac{n_h}{n_h - 1} \sum_{i=1}^{n_h} (z_{hi} - \bar{z}_h)^2$$

Don't worry — we'll build intuition before diving into formulas!

Preparation for Day 3

To Get the Most from Tomorrow:

1. **Complete the Spatial Challenge**
   - Practice will reinforce today's skills
   - You'll need spatial data tomorrow too
2. **Review Lab 2.1–2.4 Scripts**
   - Understand the survey design setup
   - We'll build on these designs for weighting
3. **Bring Your Questions**
   - Weighting challenges from your NSO?
   - Variance estimation puzzles?
4. **Optional Reading**
   - Lumley (2010), Chapter 7: Post-stratification
   - Lohr (2022), Chapter 8: Nonresponse

Day 2 Complete

Computational Tools Mastered

R & STATA • PPS Sampling • GIS Integration
Spatial Analysis • AI/ML Applications • Field Optimization

Tonight:

- Complete Spatial Challenge
- Optional: Evening lab (19:00)

Tomorrow (09:00):

- Map Gallery (15 min)
- Day 3: Weighting begins!

Thank you! See you tomorrow!